Navigation Technologies

CEC 300: Computing in Aviation and Aerospace

Authors: Richard S. Stansbury

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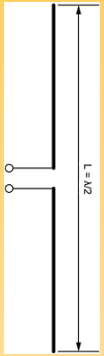
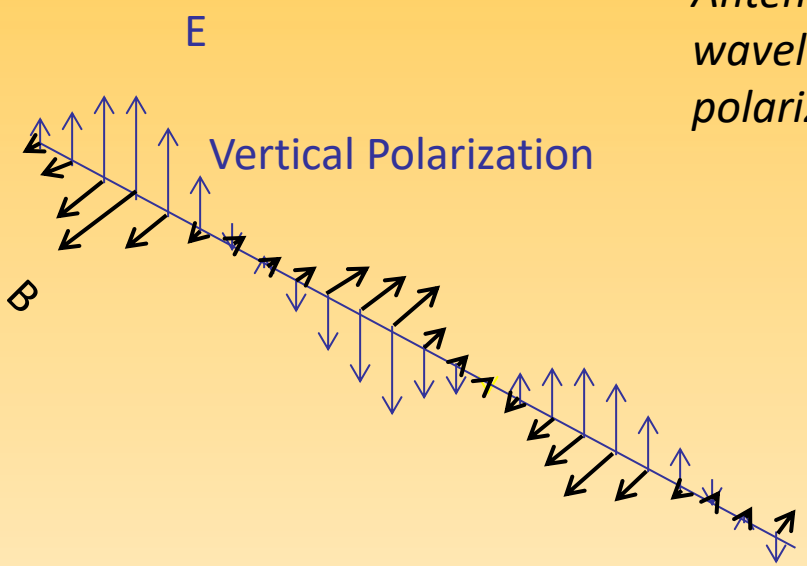
Aeronautical University

Polarization and Wavelength

Antennas work best when 'designed' for the wavelength of the signal, and has proper polarization

Transmit Antenna

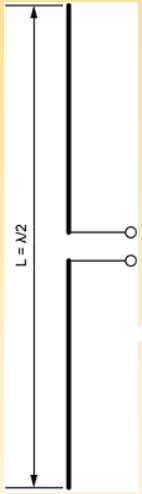


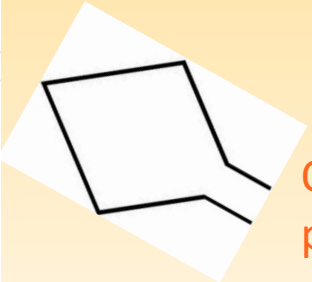


Vertical Polarization

Optimal ***dipole antenna length*** perimeter is around half wavelength (λ/2)

Electrons in antenna are "jiggled" by passage of electromagnetic wave





Optimal ***loop antenna*** perimeter is around one wavelength (λ)

Receive Antenna

RF Concepts – Antenna Ground planes

- A typical wire antenna length is around ½ a wavelength.
- If you locate an antenna above a ground plane, a mirror image of the antenna appears. This means we can use a ¼ wavelength antenna.
- The ground plane requires the ground (soil) to be conductive.
- Antennas have dozens of radial wires buried in the ground to provide a good ground plane.

RF Concepts: Electrical Length

- Even with a ground plane it is often infeasible to have a ¼ wavelength antenna.
- Adding capacitance to the end of the antenna makes the antenna look longer to electrical signals.
- Capacitive “hats”

Wavelength

- Assume a radio operates at 300kHz. What is the wavelength of the transmitted signal?
  - Wavelength = Speed of light/frequency
  - $\text{Wavelength} = \frac{3\times10^8\ m/s}{3\times10^5\ cycles/s}$ = 1000m
  - Efficient antennas are ¼ of a wavelength.
  - The antenna needs to be 250m tall?

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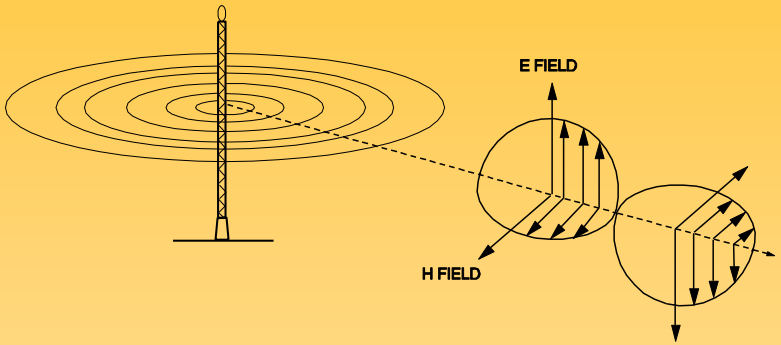
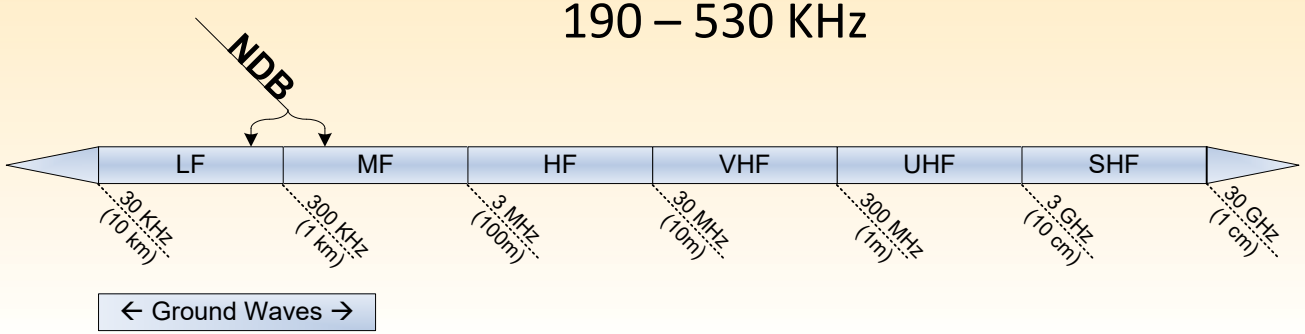
NON-DIRECTIONAL BEACONS

A Non Directional Beacon, NDB



- Transmits vertically polarized waves
- Uses Ground Waves which follows the curvature of the earth.
- Identify with morse code.

Propagates by *Ground Waves*
190 – 530 KHz



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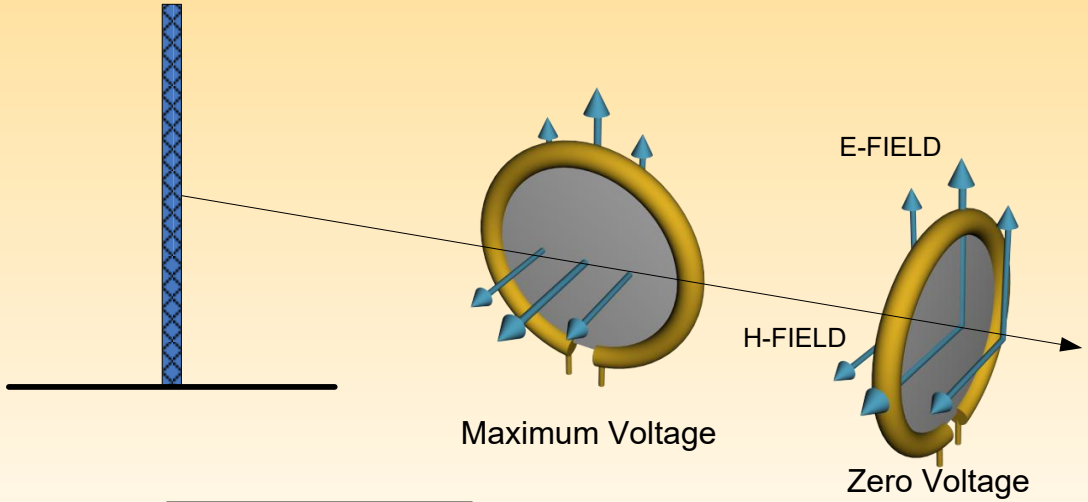
A Typical NDB Station

Source: G4FTC

Direction Finding Loop and



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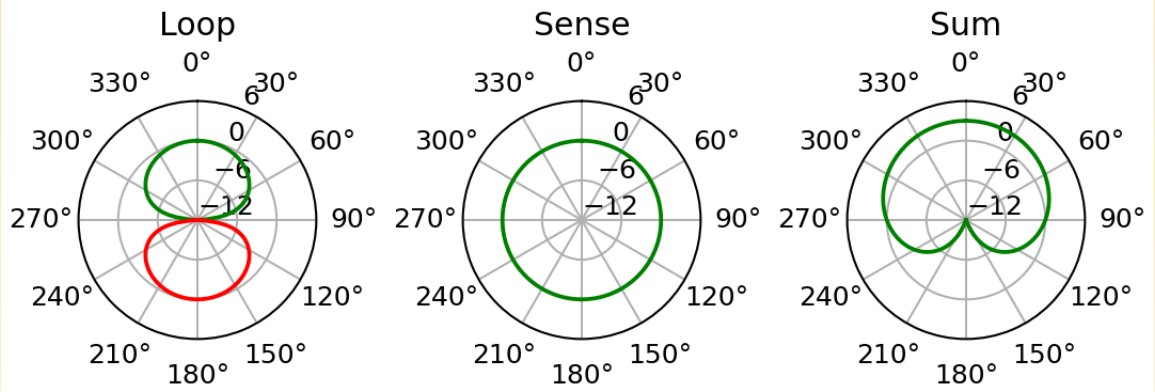
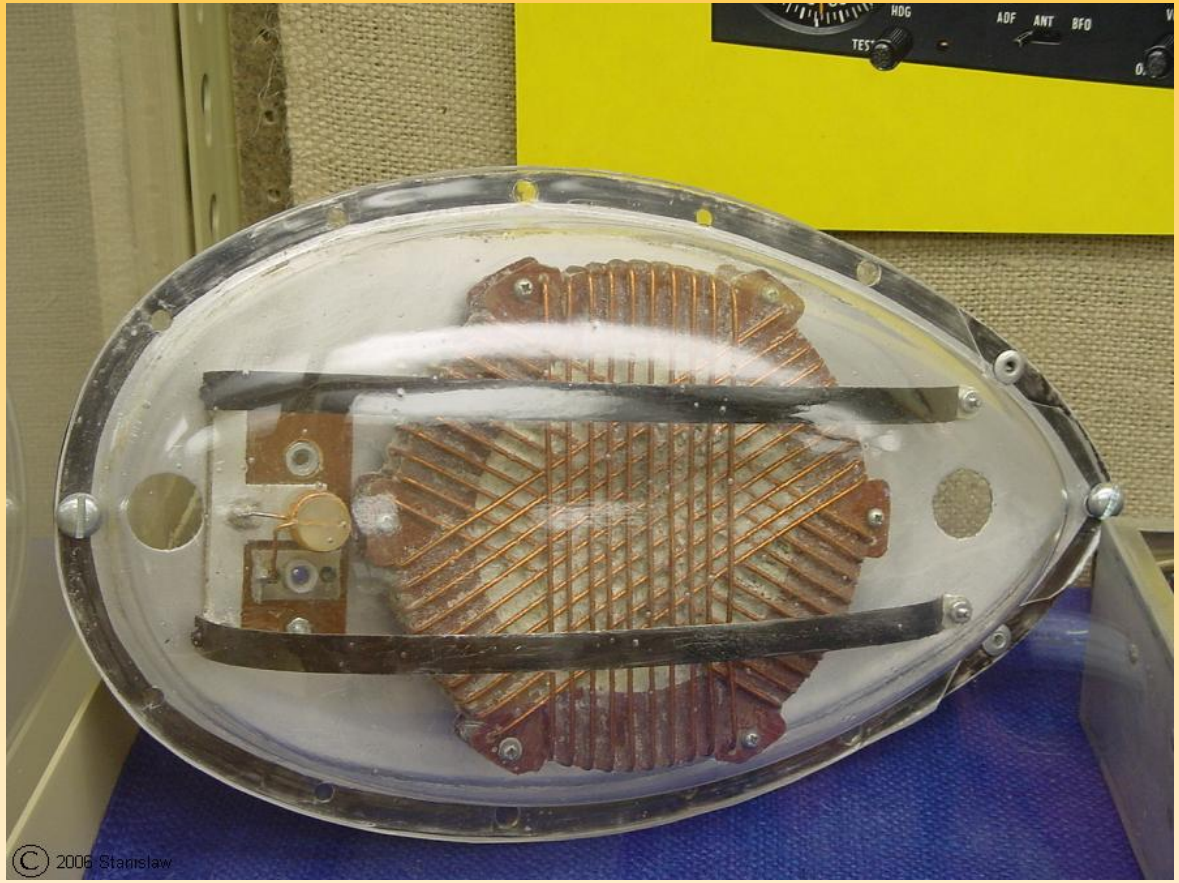
Two Antennas: Loop + Sense

A single steerable loop can work, but it is harder to identify zero.

The addition of a sense antenna with the loop antenna is such that when the antenna is pointed away from the target, the receive power goes to zero. The beacon is at the angle + 180 degree.





Solid state antenna replaces loop antenna

NDB: Frequencies

- ICAO allows operation from 190kHz – 1750kHz.
  - In many countries 540 -1610kHz is used for commercial AM radio broadcasts
- In the US NDBs operate from 190 kHz – 435 kHz and 510 kHz – 530 kHz
- 500kHz used to be a distress frequency (now unused)

Shore Effect

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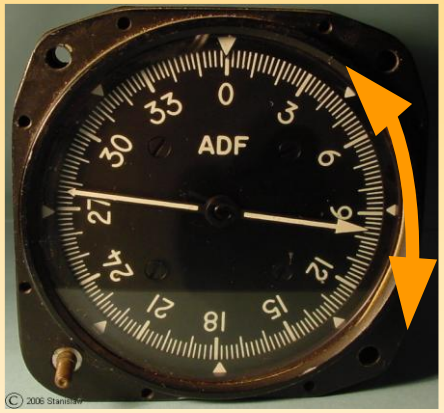
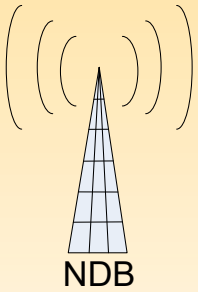
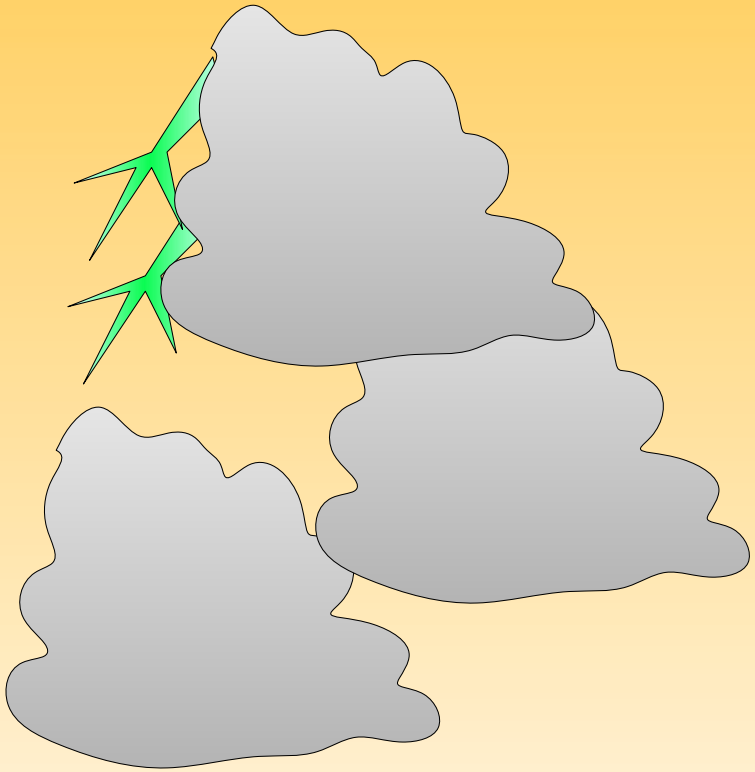
Precipitation Static







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A Dual Pointer Radio Magnetic Indicator, RMI

VOR – VERTICAL OMNIDIRECTIONAL RANGE

Bearing Systems

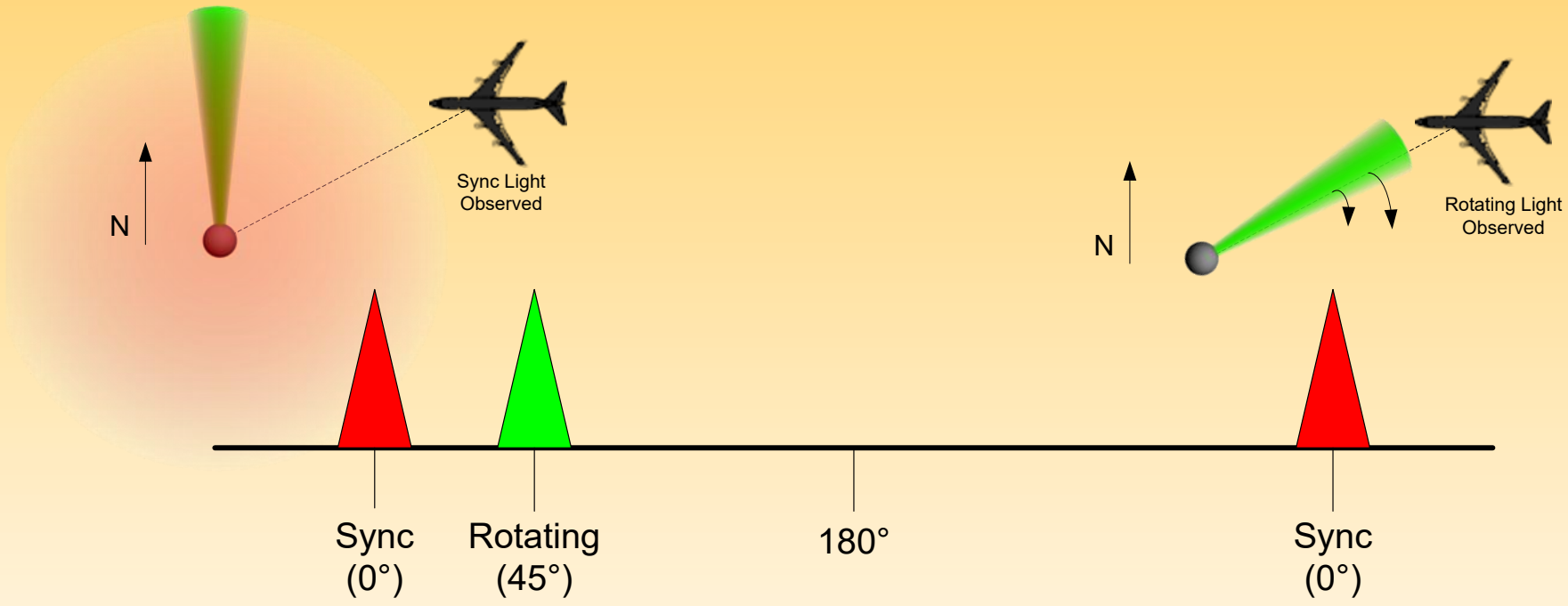
- General Principle:

- Need two signals:
  - One whose phase varies with bearing from (or to) the transmitter

  - 2. A reference signal whose phase is constant regardless of bearing

- Lighthouse example

Light Beacon Analogy



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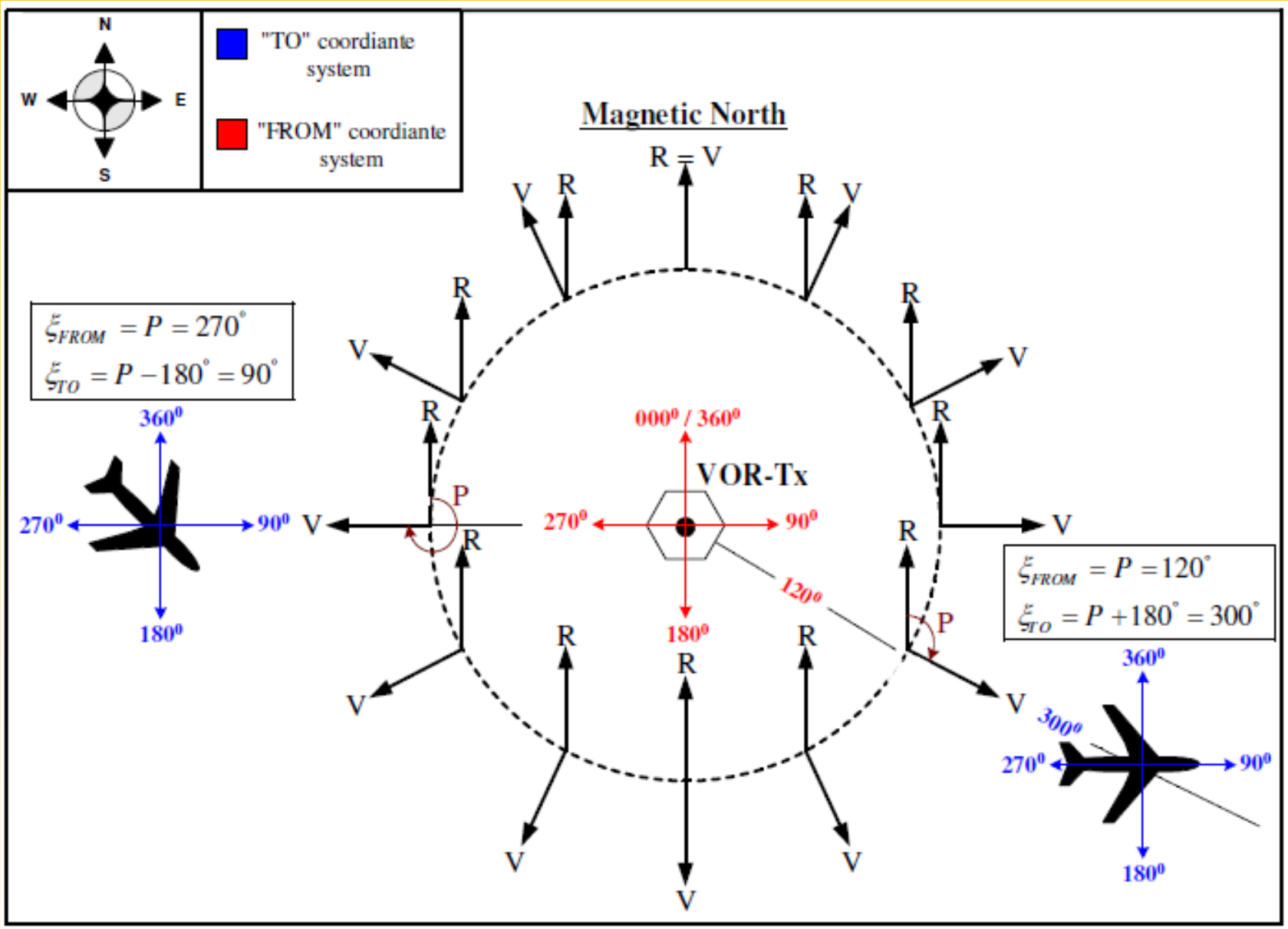

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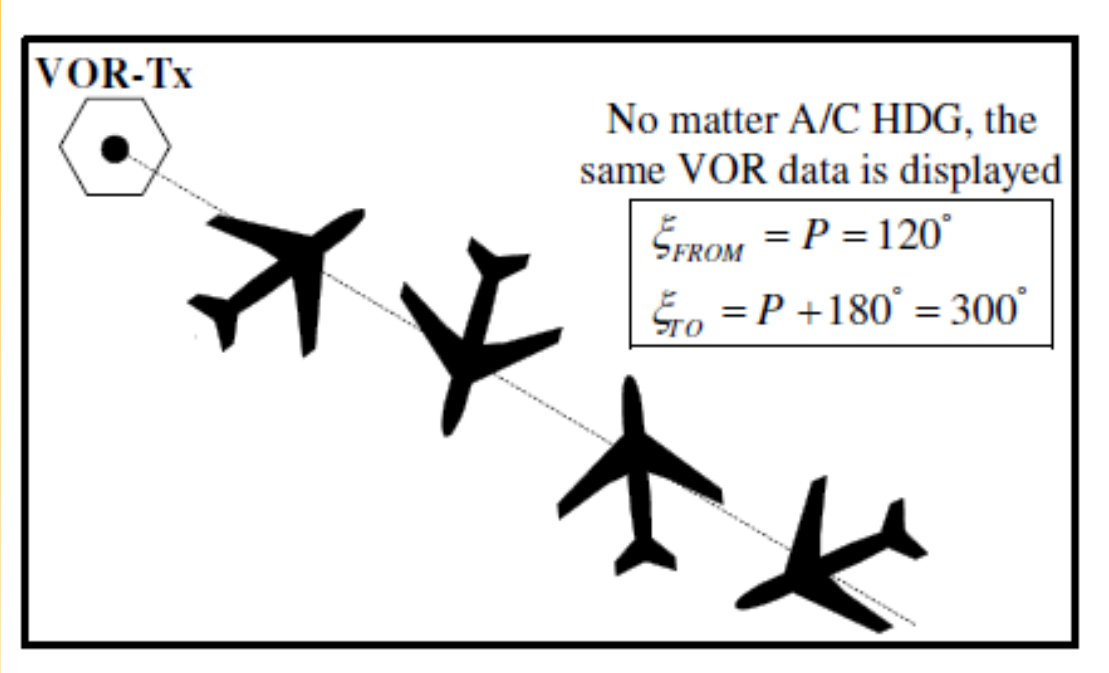
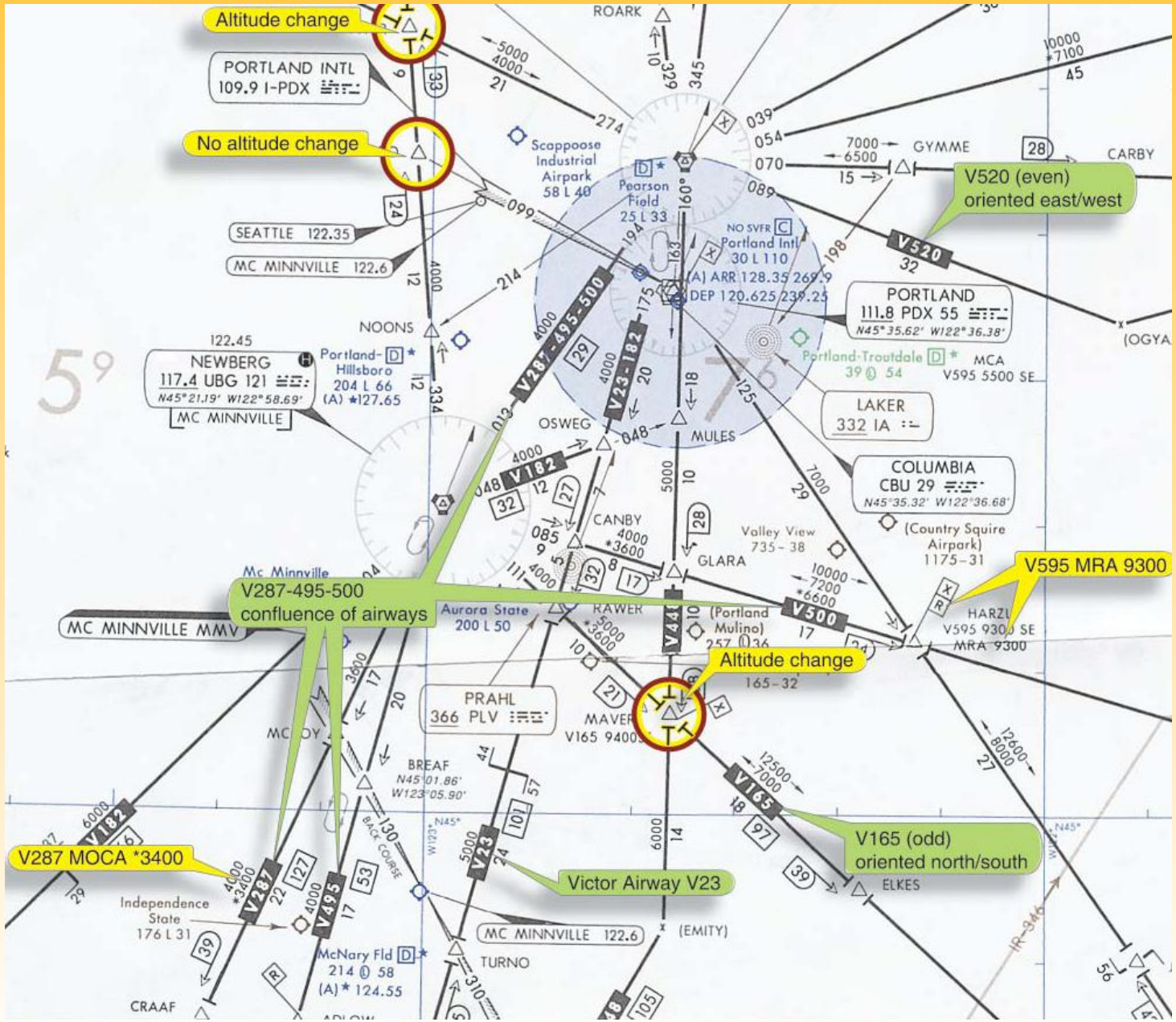
Radial, not bearing

- VOR will indicate which radial the aircraft is on relative to the station
- If the aircraft is on the radial and the radial leads toward the station, then the TO flag is set.
- If the aircraft is on the radial and the radial segment leads away the station, then the FROM flag is set.
- Heading of the aircraft is irrelevant:
  - If you are on the TO radial, but flying away from the station, it would still read as “TO”, but the pilot must reverse their controls when following the CDI.



Some Airways Based on VORs



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VOR

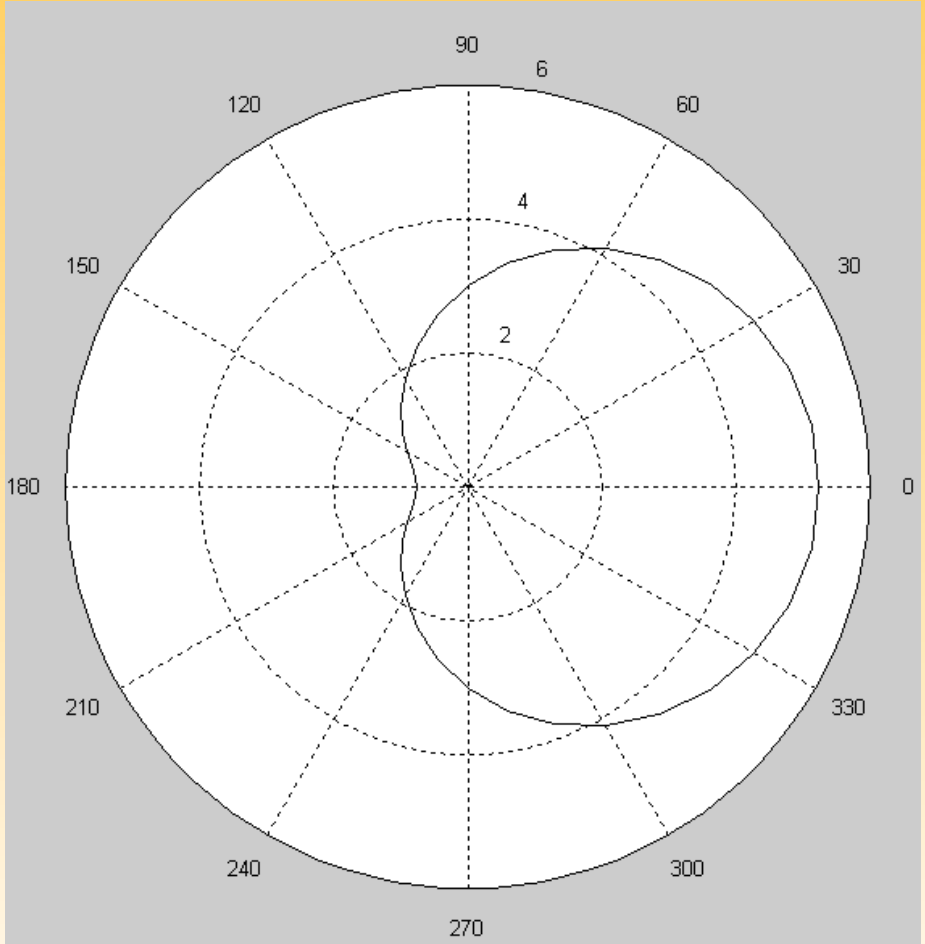
- In VOR, the variable signal is provided by a limacon antenna pattern which rotates clockwise at 30Hz.

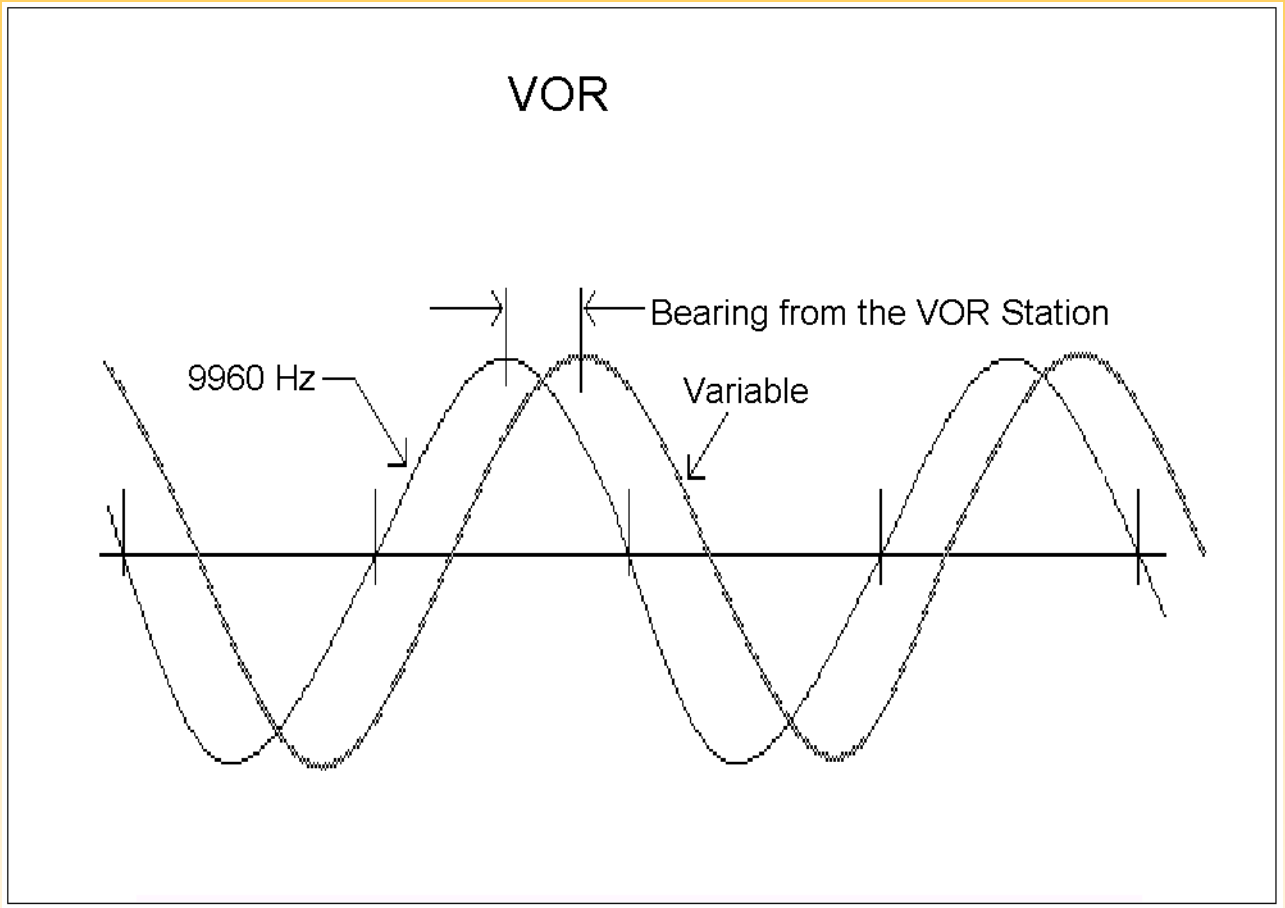
Note: a limacon has the equation

$$r = a + b \cdot \cos(\theta)$$

Thus an observer at any point measures an RF signal , amplitude modulated at 30 Hz

The Phase Angle Between the Reference and Variable is the VOR Bearing





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Variable 30 Hz & Reference 9960 Hz

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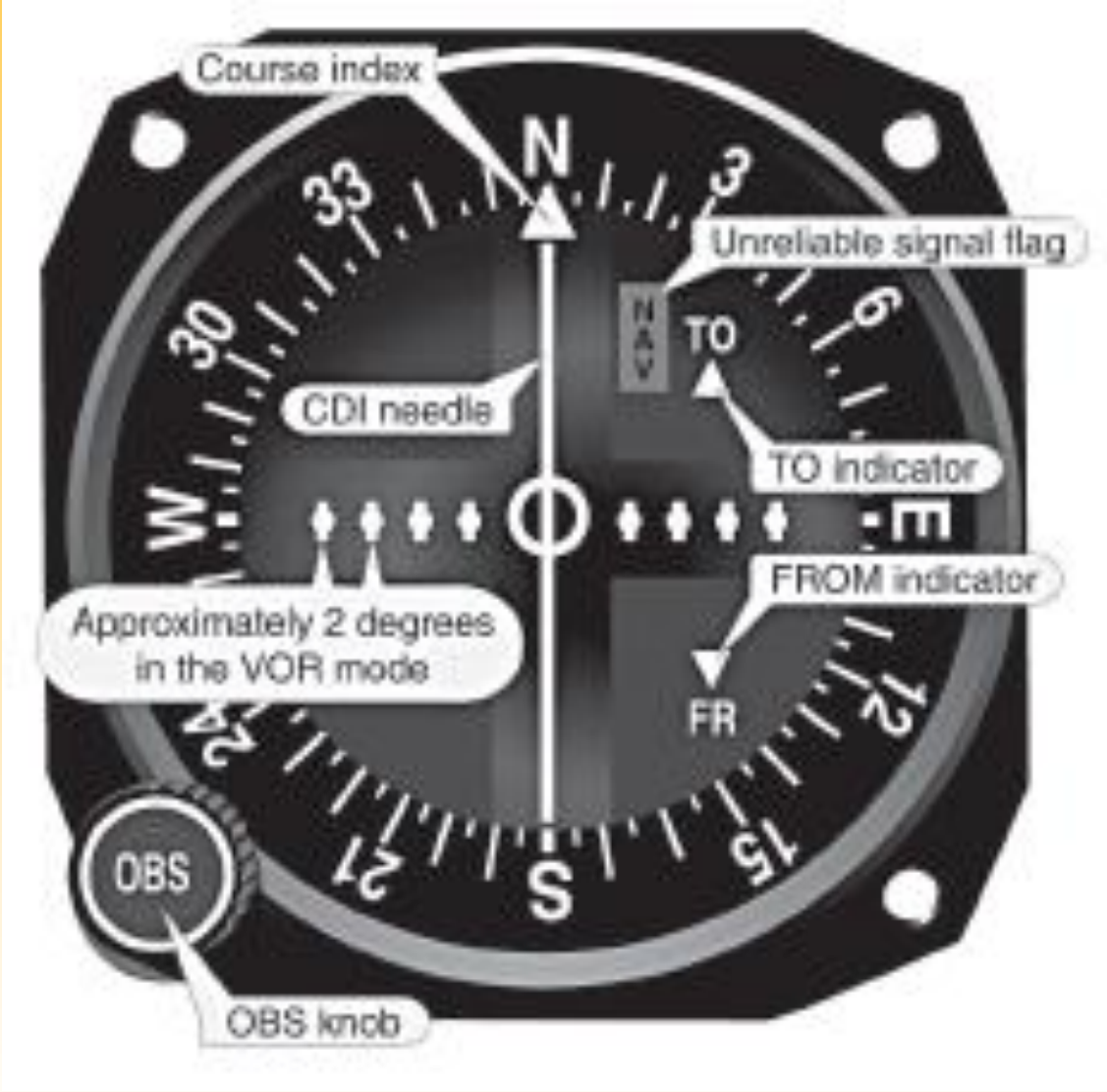
The Basic VOR Indicator



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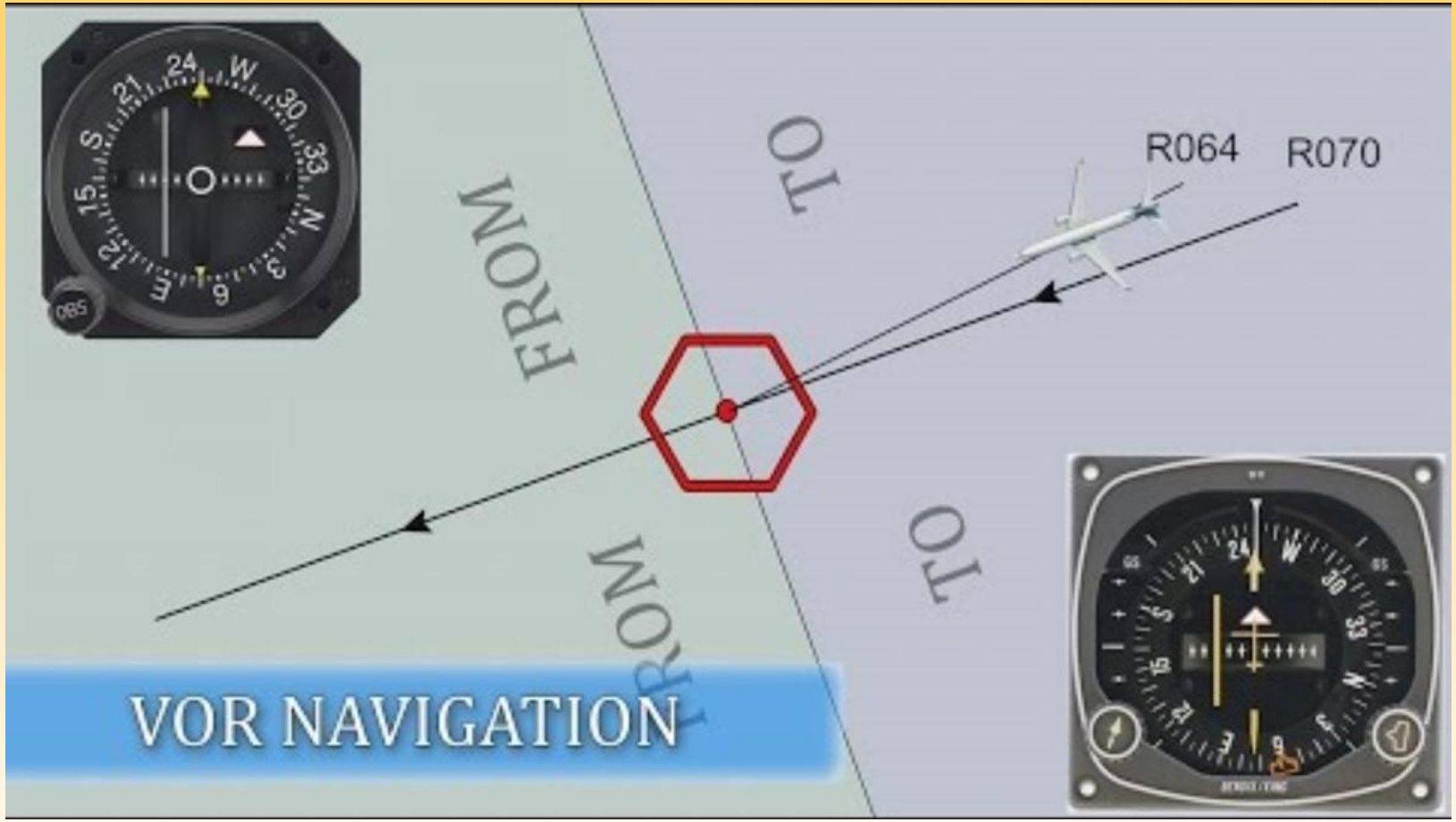
Video Demo



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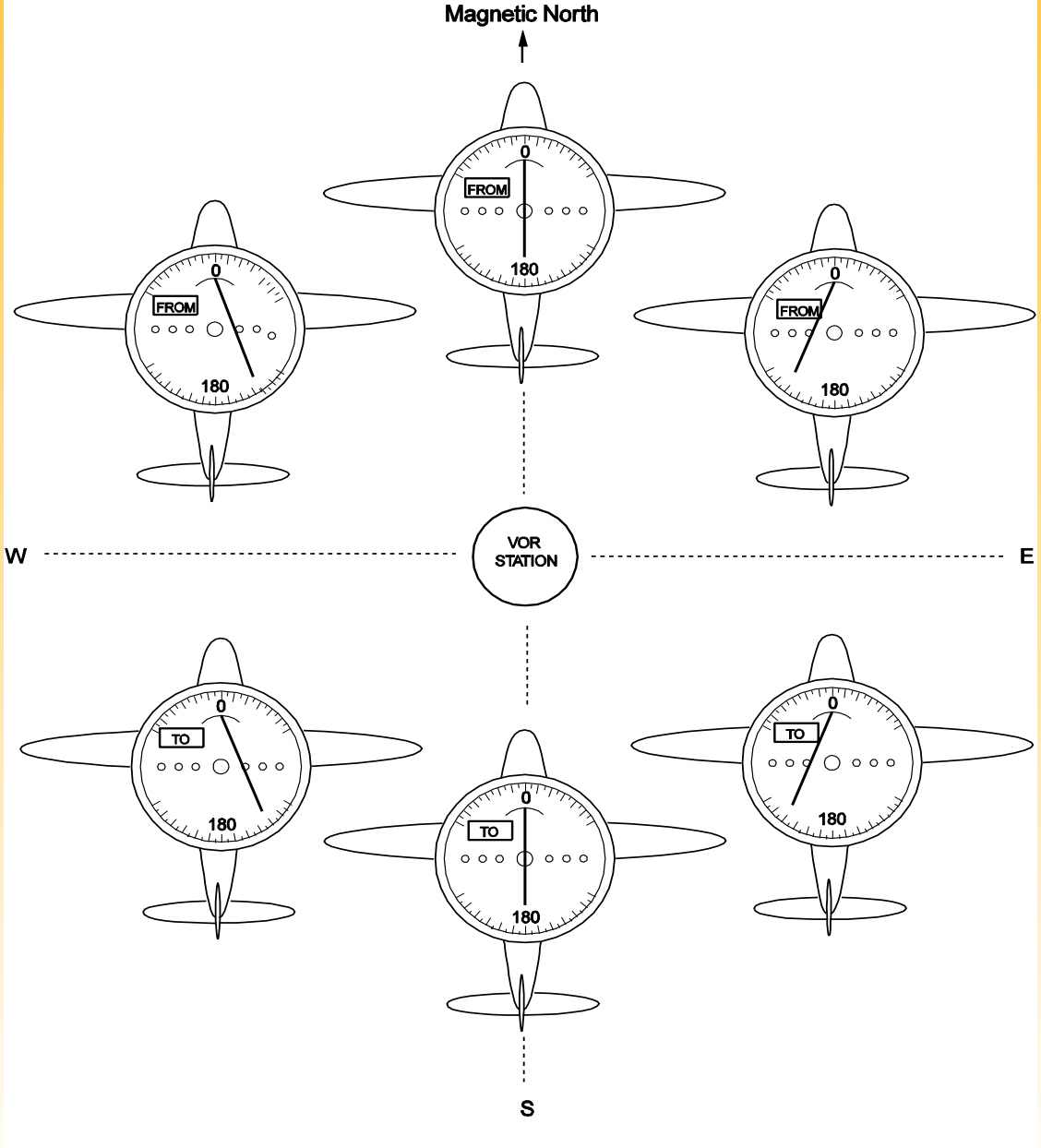
Operation of the TO/FROM



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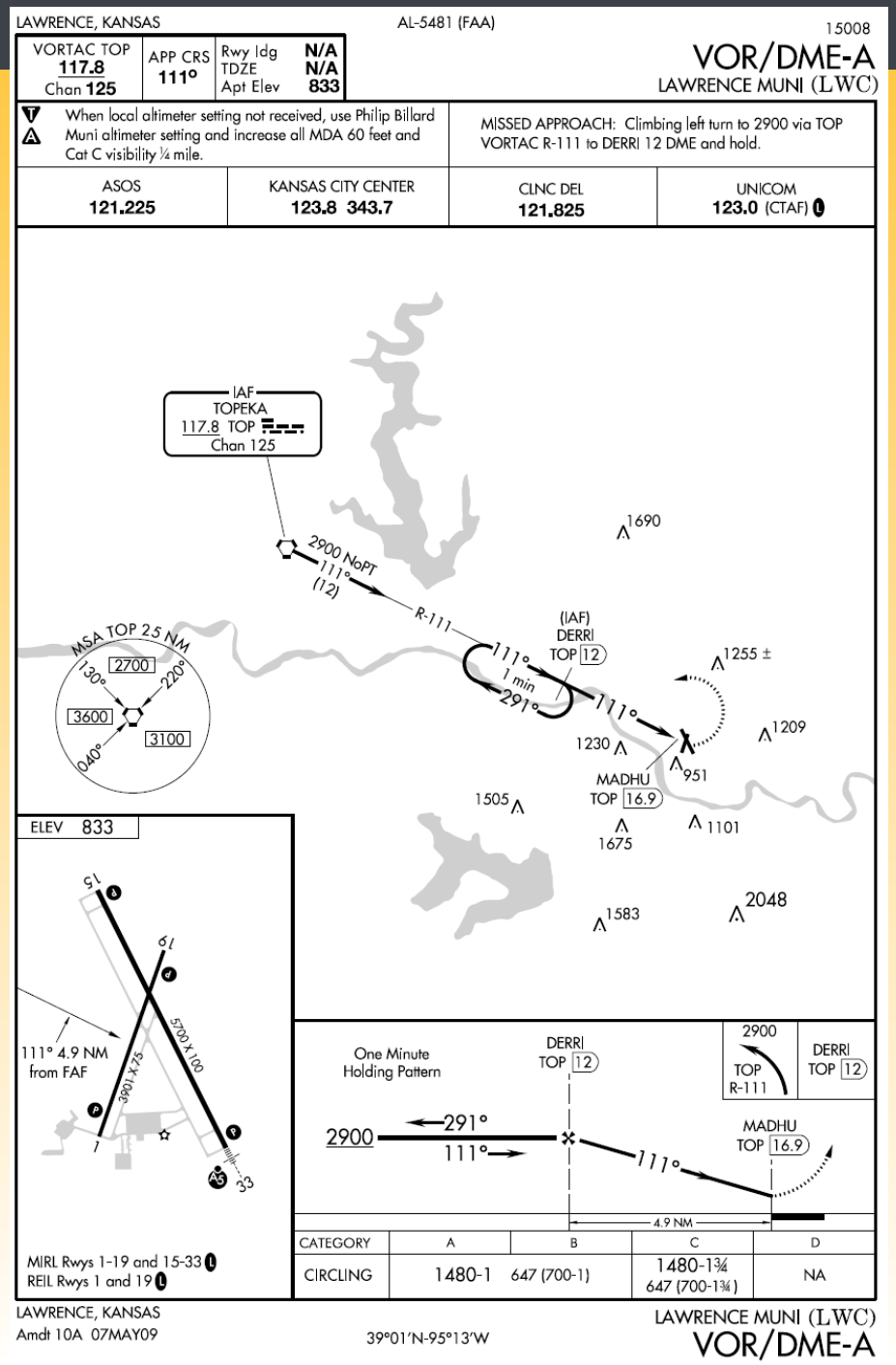
VOR Landing Aid



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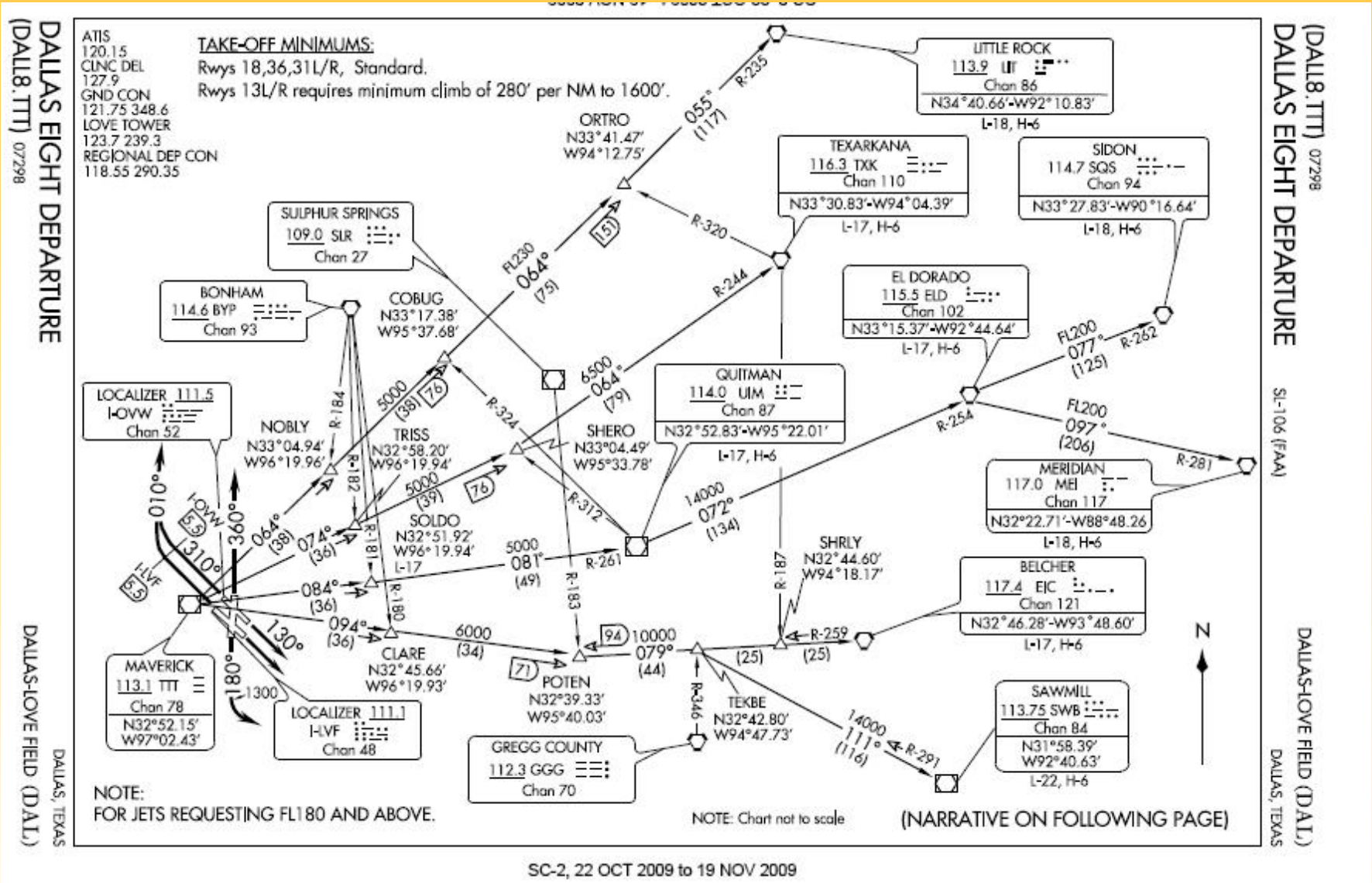
VOR Departure



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VOR Departure

DEPARTURE ROUTE DESCRIPTION

All aircraft maintain 3000 feet and expect to cross CRG VORTAC at or above 4000 feet. Expect to cross SAWGY INT at or above 12,000 feet. Expect clearance to requested altitude/flight level 10 minutes after departure.
TAKE-OFF ALL RUNWAYS: Cleared as filed. Climb runway heading or as assigned for vectors to join CRG R-140 to SAWGY INT. Then via OMN R-355 to OMN VORTAC.

SAWGY ONE DEPARTURE
(SAWGY1.OMN) 07298

JACKSONVILLE, FLORIDA
JACKSONVILLE INTL (JAX)

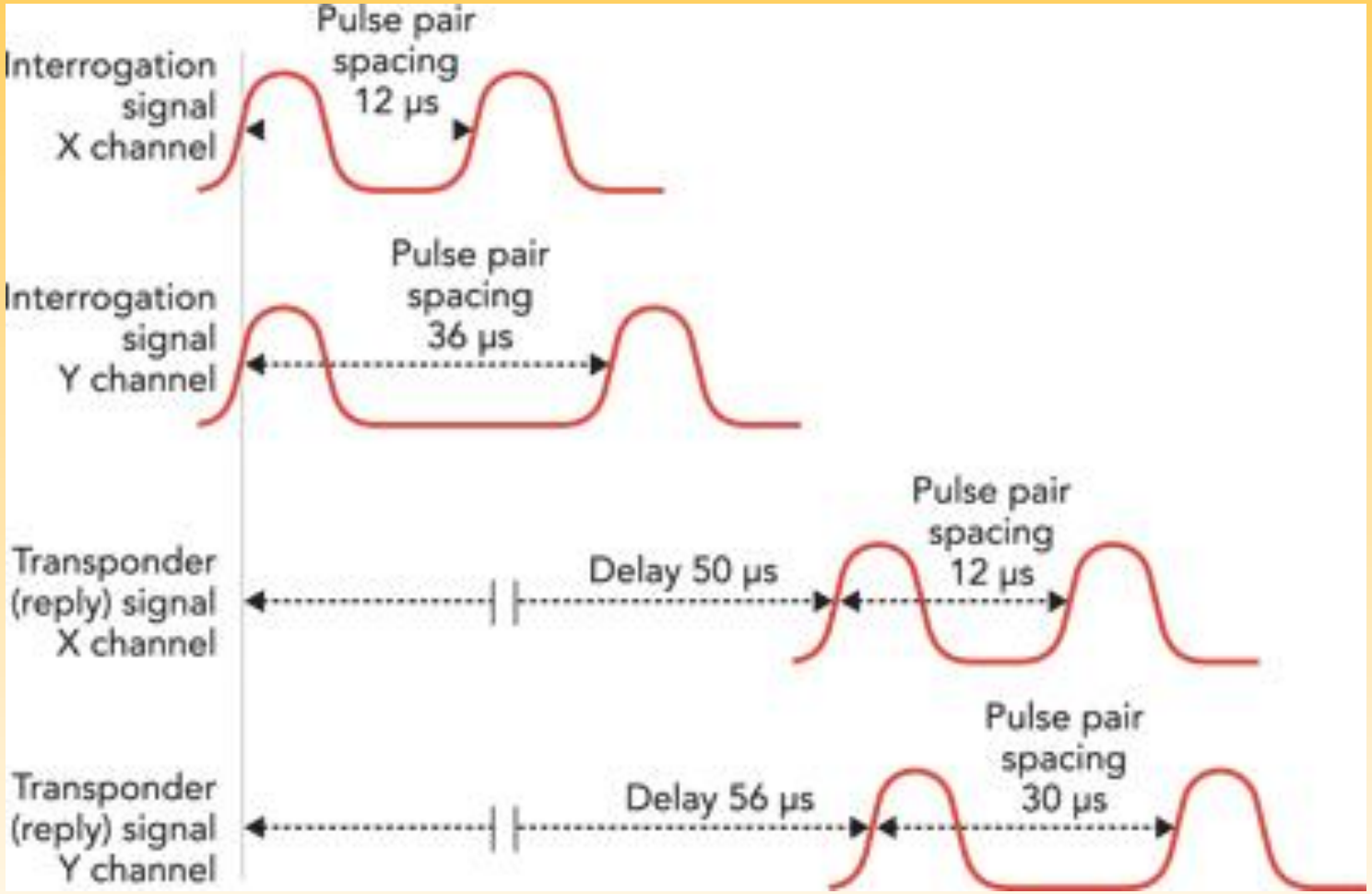
DISTANCE MEASUREMENT EQUIPMENT (DME)

DME

- Frequencies:
  - Airborne: 1025 MHz – 1150 MHz
  - Ground :
    - 63 MHz below Tx frequency 1025 – 1087 MHz
    - 63 MHz above Tx frequency 1088 – 1150 MHz
- System sends pairs of pulses with the following spacings.

Channel	Delay in Ground (Reply pulse pair spacing)	Delay in Air (Interrogation pulse pair spacing)
X	12us	12us
Y	30us	36us
W	24us	24us
Z	15us	21us

Pulse pairs


Interrogation signal X channel
Pulse pair spacing 12 µs
Interrogation signal Y channel
Pulse pair spacing 36 µs
Transponder (reply) signal X channel
Delay 50 µs
Pulse pair spacing 12 µs
Transponder (reply) signal Y channel
Delay 56 µs
Pulse pair spacing 30 µs

Radar Mile

- A radar mile is the time required for a signal to travel out to a target 1 nautical mile away and return.

- Time= 2 x distance/speed of light

- Radar Mile = 2 x 1852m/3x10^8 m/s = 12.3 microseconds

DME distance

- The DME ground station adds a fixed delay of 50 microseconds before retransmitting the pulse (for X channel).
- Distance in nm = (time – 50 microseconds)/12.3 microseconds
- The maximum range allowed for use in FARs is 200nm.
- 200nm*12.3microseconds+ 50 microseconds = 2510 microseconds
- DME can be used to LOS which can be up to 300nm.

FRUIT

- There can be a lot of aircraft interrogating a DME transponder.
- Many of the pulses coming back to the aircraft are responses to DME pulses from other aircraft.

- These replies are known as Friendly Replies Unsynchronized in Time.
  - Commonly called FRUIT.

- The circuit to remove these unwanted replies is a deFRUITer.

Garble

- A valid pulse pair for the X channel will have a pulse spacing of 12 us.
- Suppose a pulse from other aircraft pulses winds up in 12us window.

- The pulse spacing is not valid for any channel and the DME will not respond to these pulses.

- Interference that scrambles the signals and prevents a correct output is called GARBLE.

Random Interrogations

- If two aircraft's pulses cause garble, the aircraft will retransmit the pulses.
  - The new pulses could garble
    - Repeat …. Forever.
- To avoid this problem the interrogations are issued with random delays.
- The average rate is 15 pulse pairs/second

SQUITTER

- SQUITTER is a response to no one.

- The transmitter and receiver have trouble if they only receive occasional pulses
- DME guarantees 2700 pulse pairs/second.

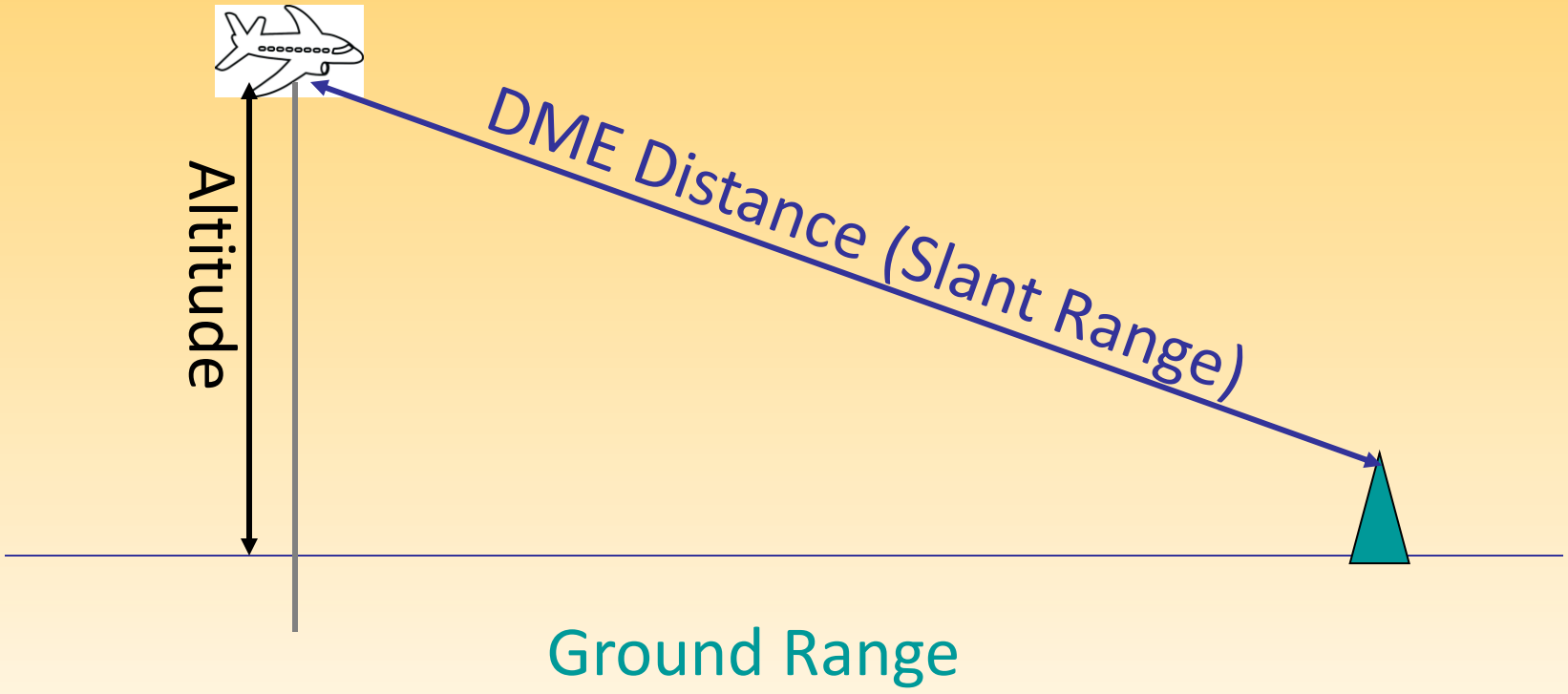
- Some of these pulses are used to transmit the Morse code identifier for the DME

- Some of the pulses are used to support TACAN (when available)

Capacity

- Out of the 2700 pulse pairs/second that are available,
- about 2000 are available to aircraft.
- 2000 pulse pairs/sec divided by 15 pulse pairs /sec for each aircraft.
- A DME can support 133 aircraft.

Slant Range



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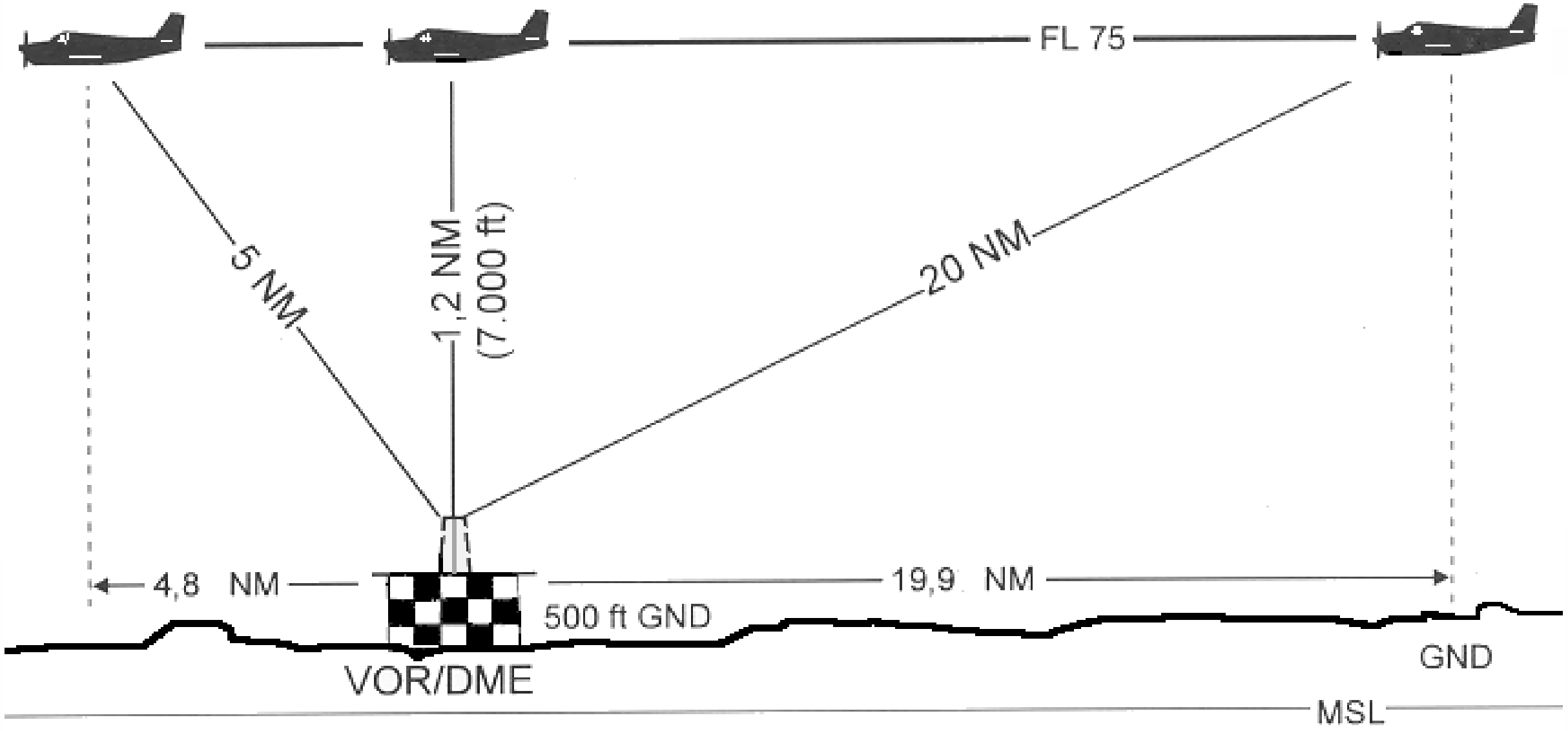
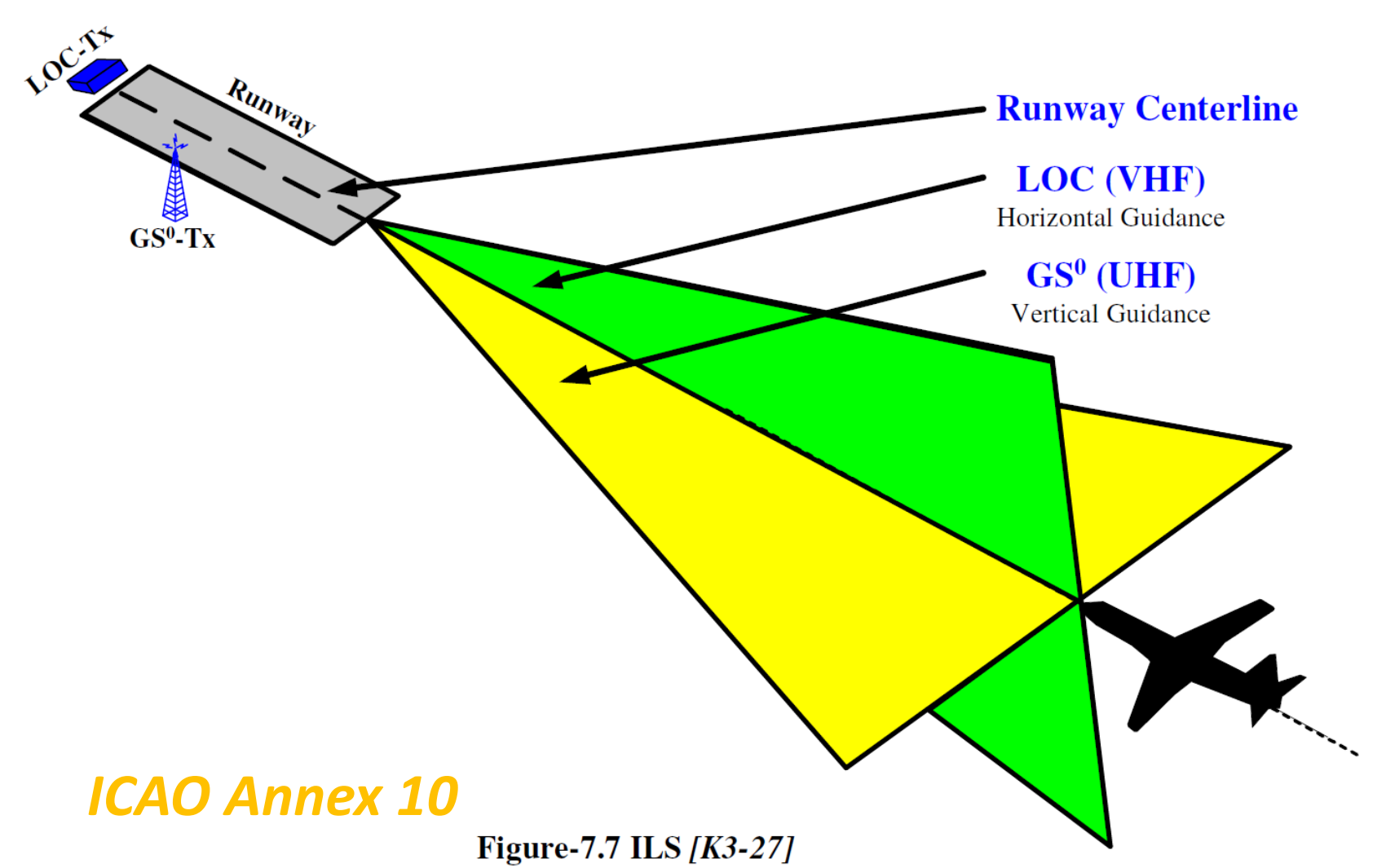

FL 75
5 NM
1,2 NM
(7.000 ft)
20 NM
4,8 NM
19,9 NM
500 ft GND
VOR/DME
GND
MSL

INSTRUMENT LANDING SYSTEM

General Concept



ICAO Annex 10

Figure-7.7 ILS *[K3-27]*

From: Avionics Made Simple by Abdulla, Svoboda, and Rodrigues.

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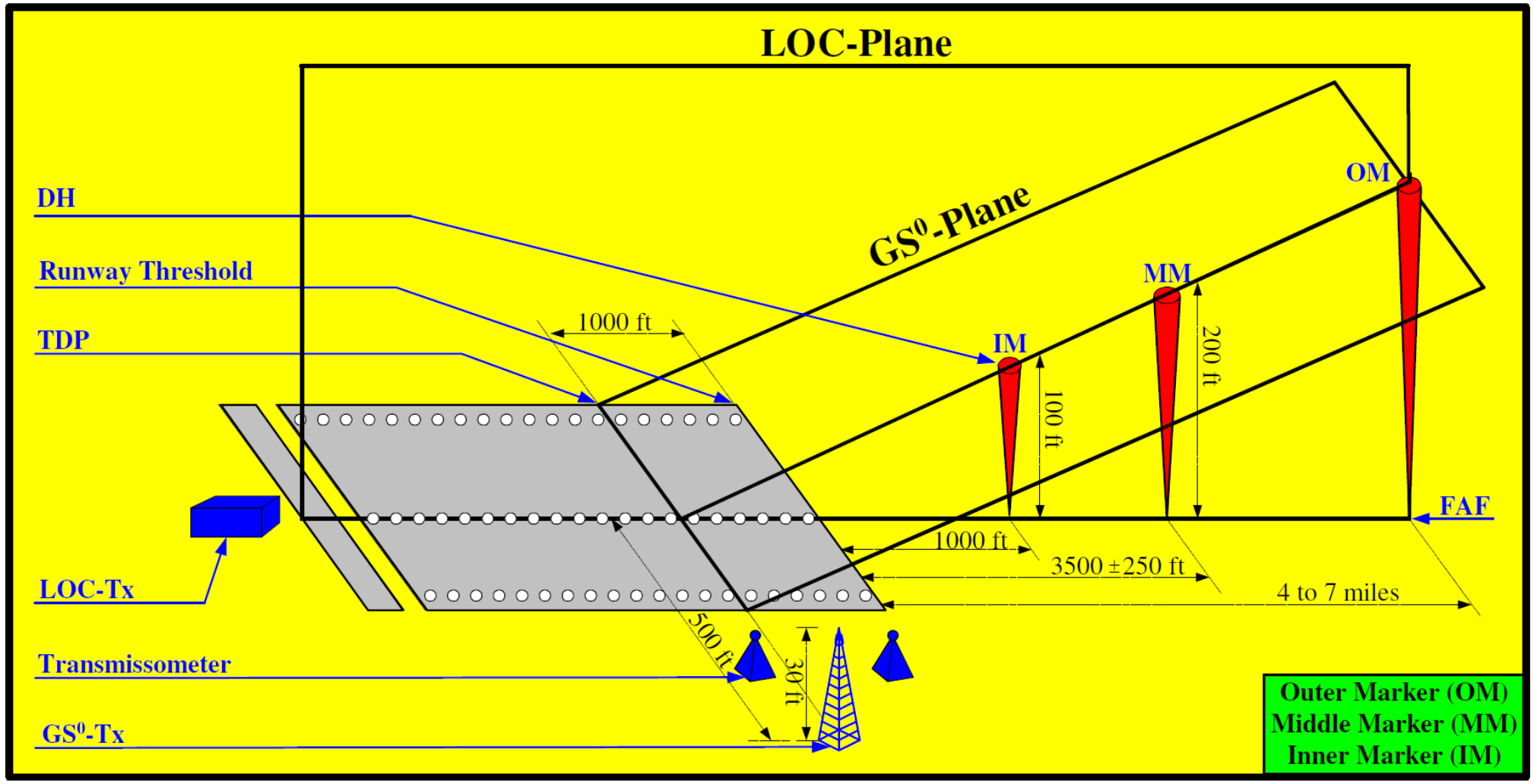
General Concept



From: Avionics Made Simple by Abdulla, Svoboda, and Rodrigues.

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Localizer (LOC)

- Provides alignment with the runway centerline, on a horizontal plane.
- The LOC transmitter is located at the end of each runway.
- Frequency: 108-112 MHz (VHF) with 20 channels.
- Range: 40 km.
- Maximum deviation from the centerline covered: +- 2^0 (4^0 total).

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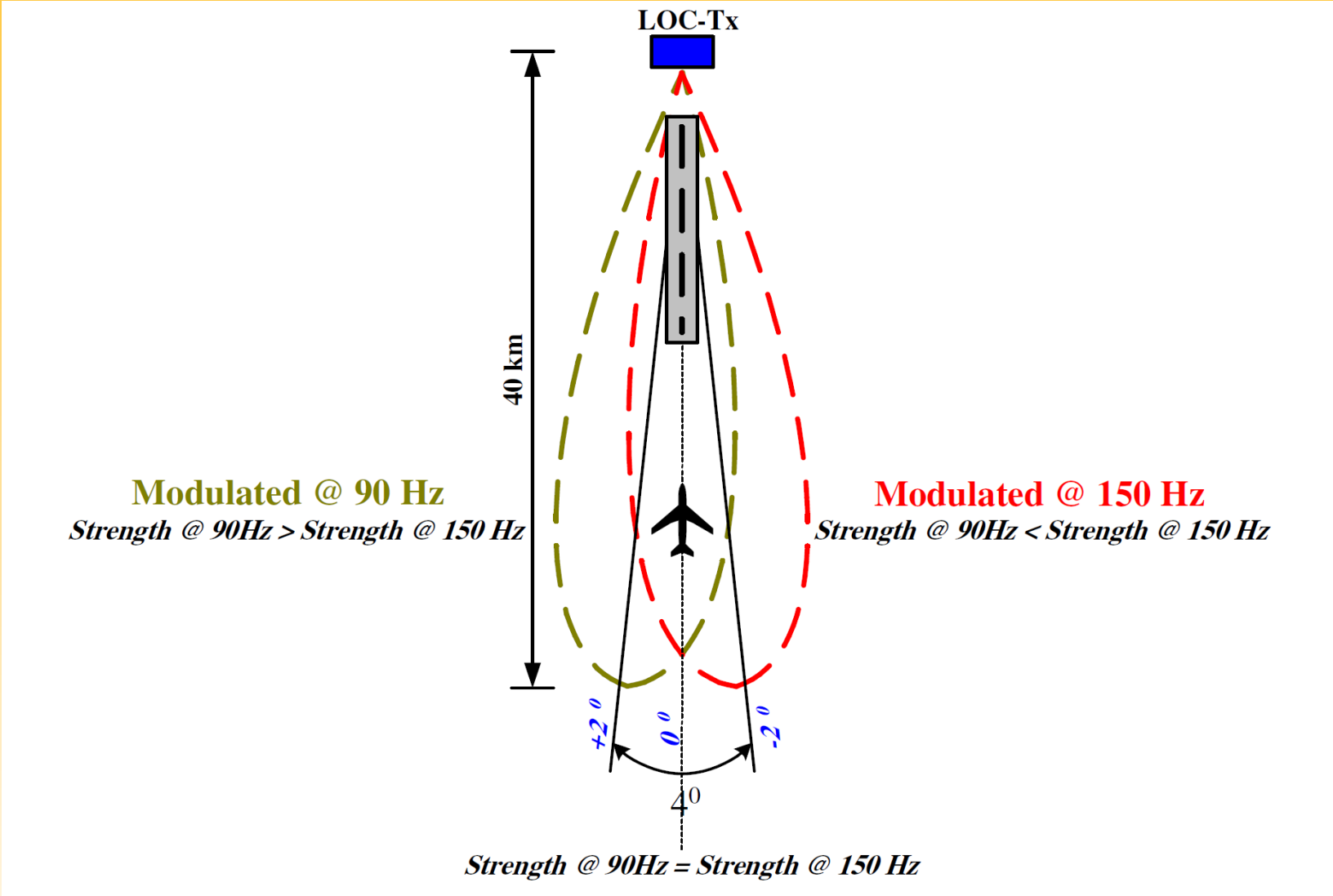
Localizer



From: Avionics Made Simple by Abdulla, Svoboda, and Rodrigues.

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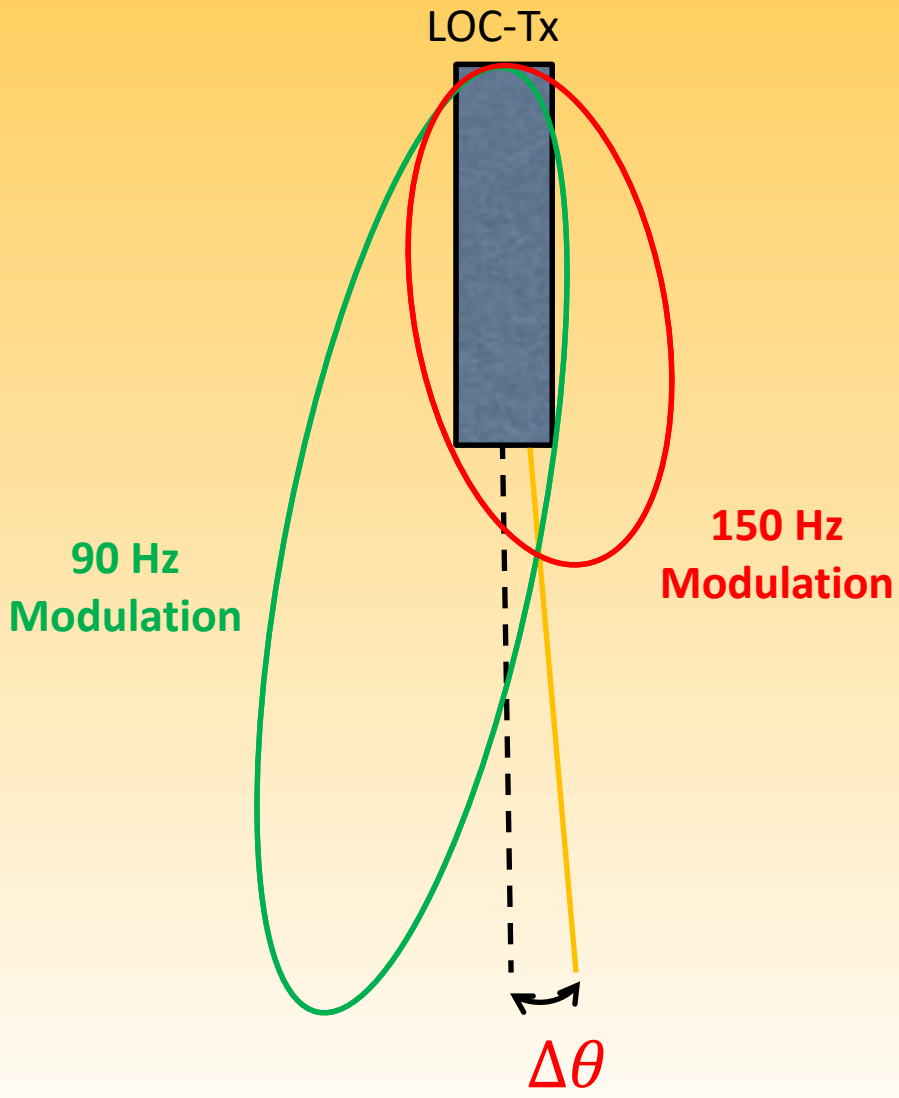
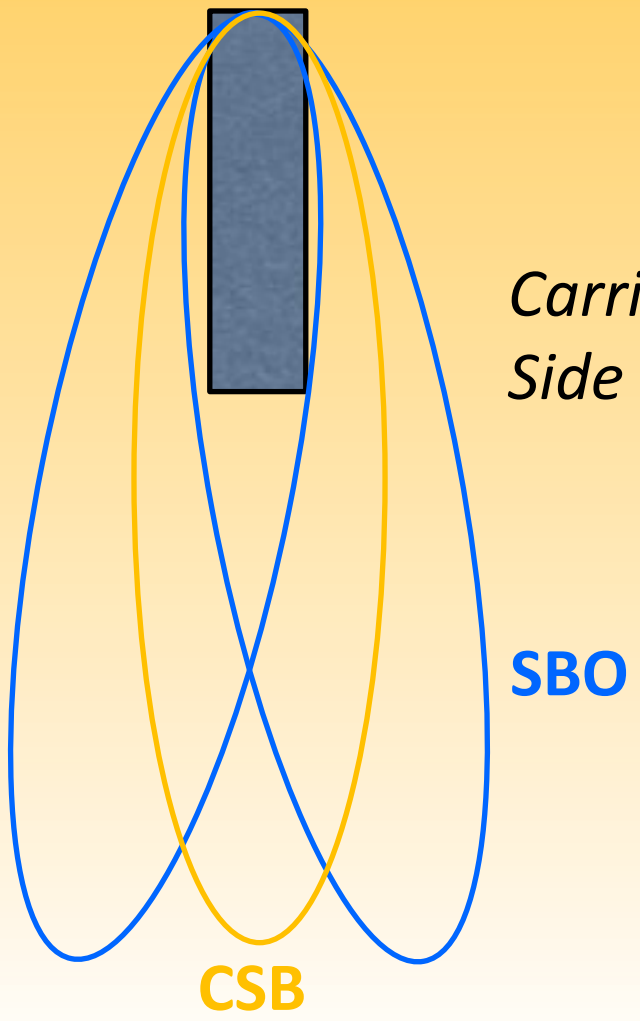
Issue: Two Antennas is not sufficient

Solution : Used two types of modulation





*Carrier with Side Band (**CSB**)*

*Side Bands Only (**SBO**)*

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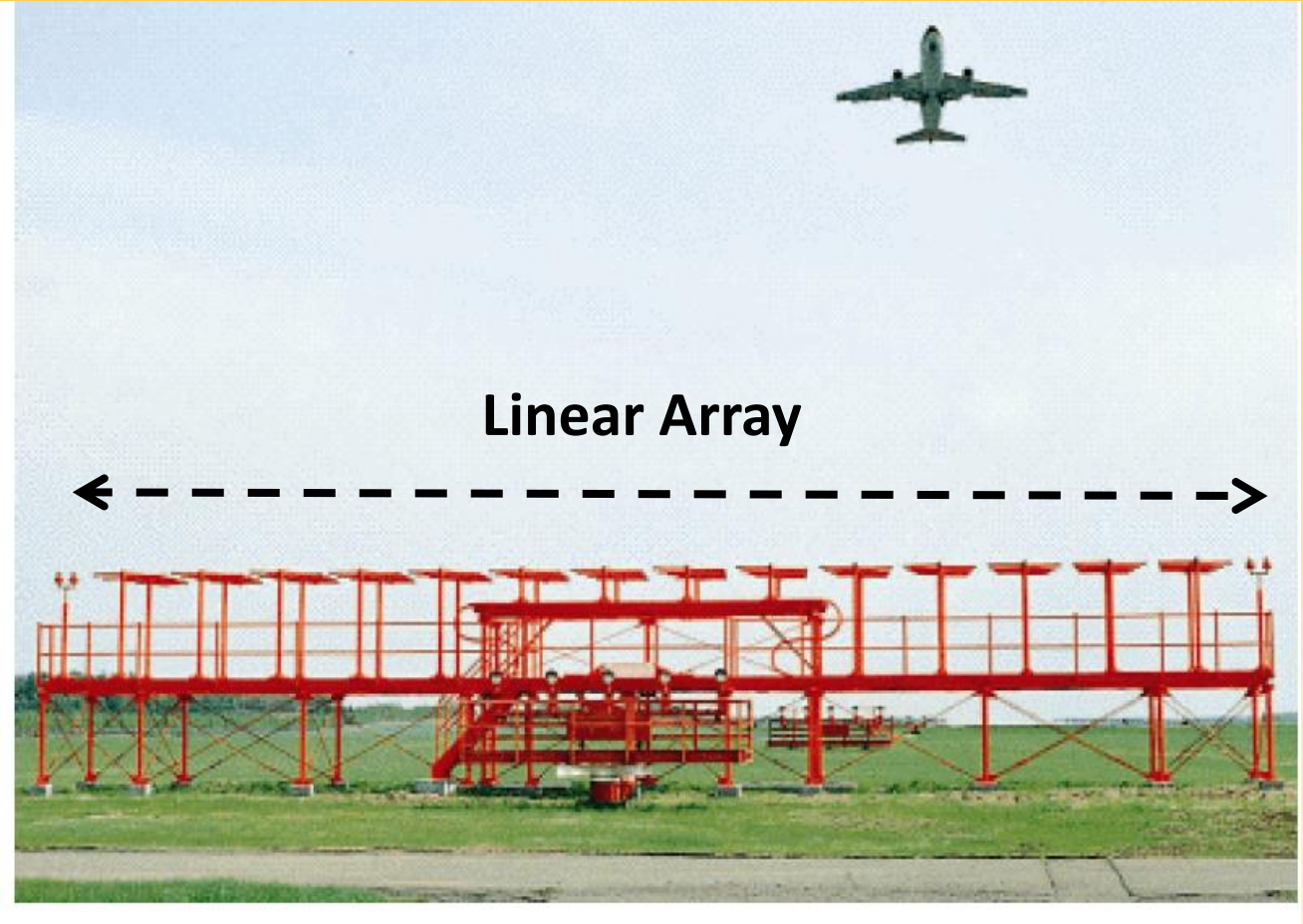
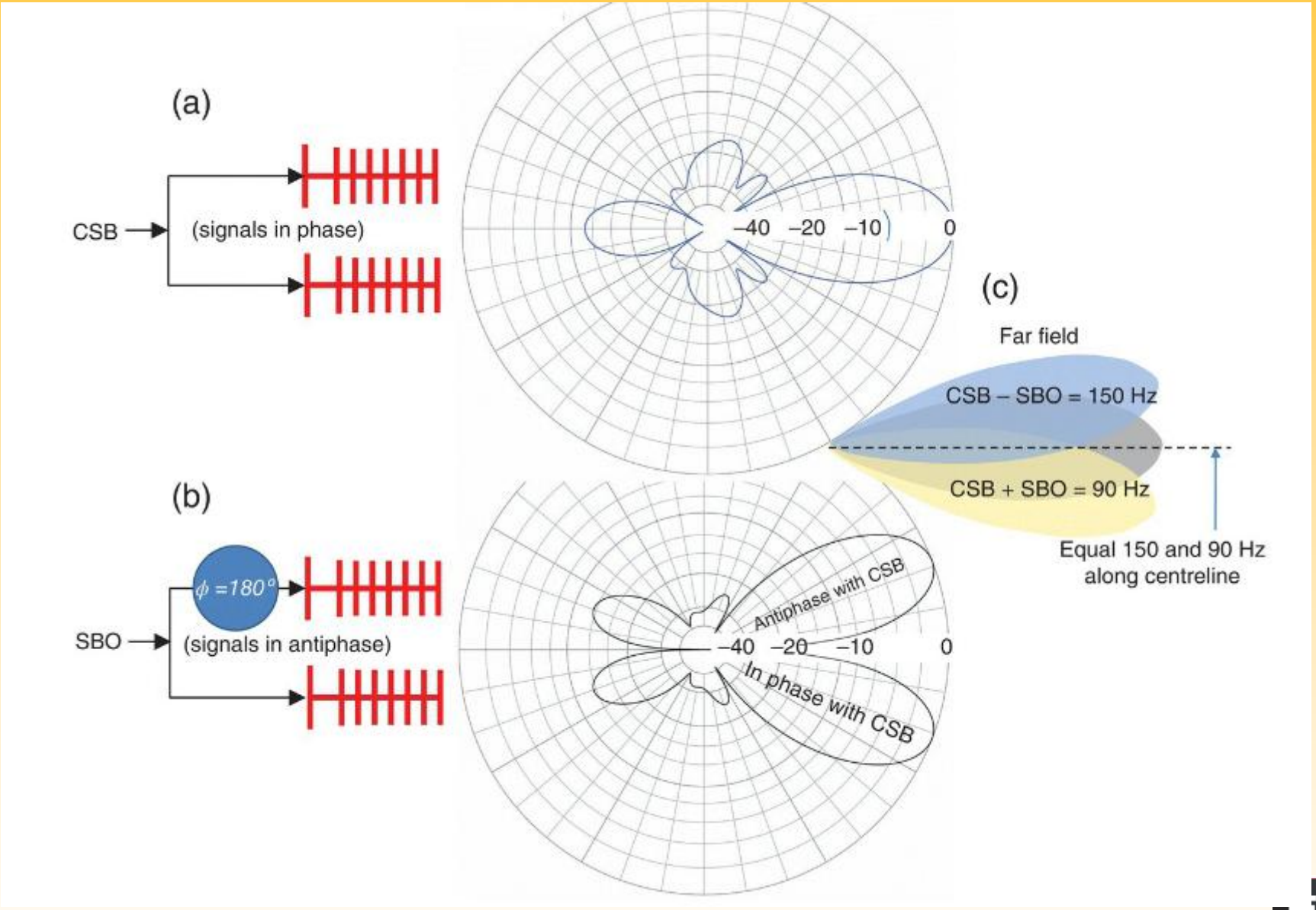
LOC-TX Antenna

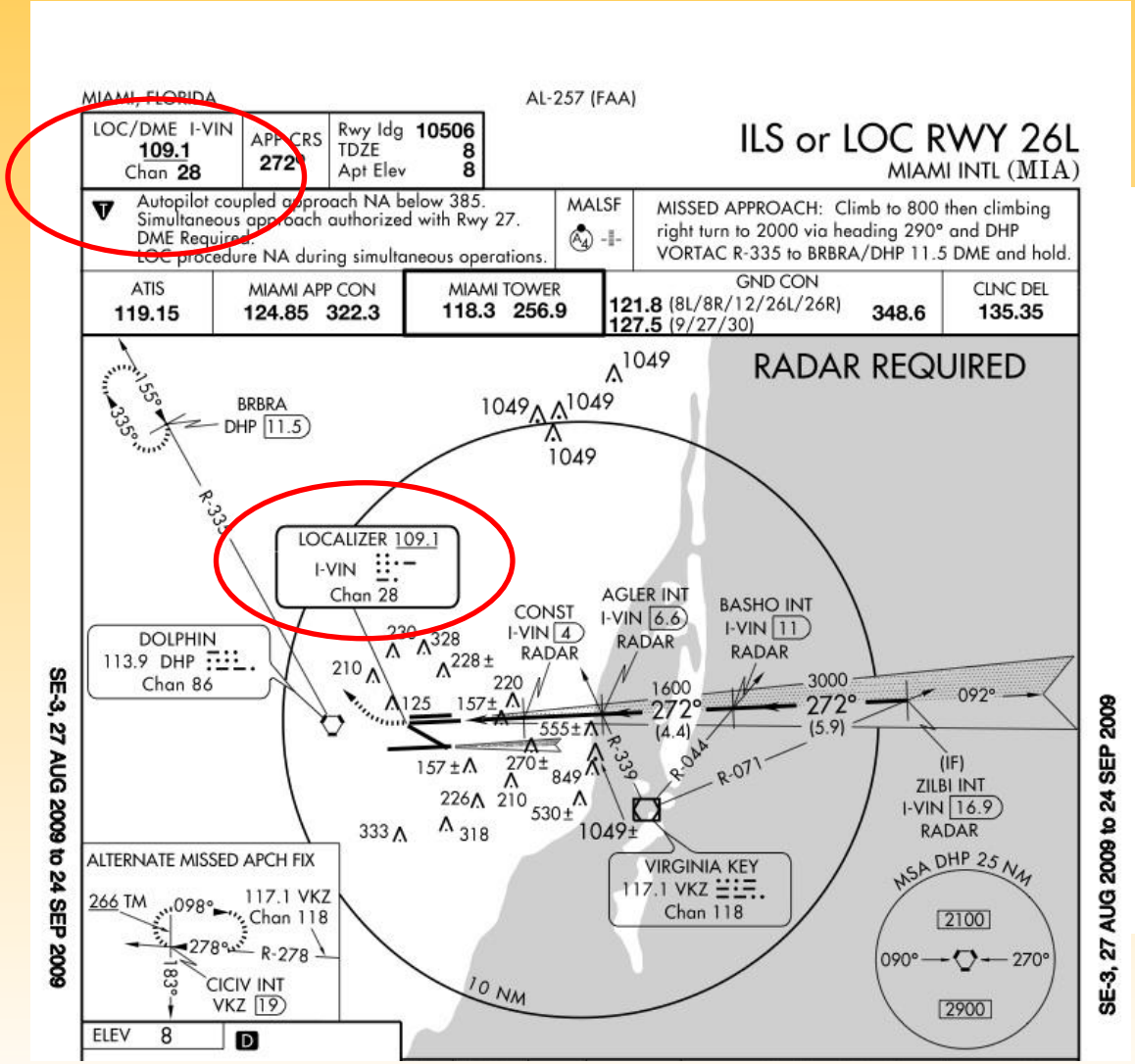


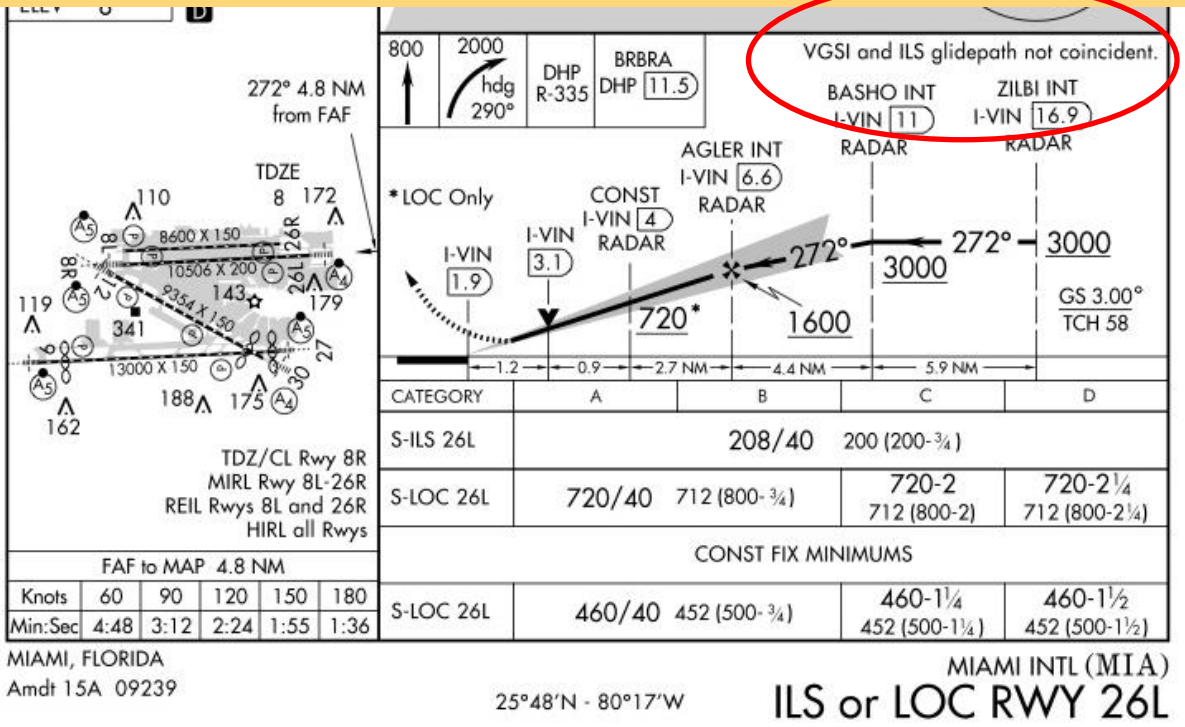
From: Avionics Made Simple by Abdulla, Svoboda, and Rodrigues.

Space Modulation



Localizer Approach





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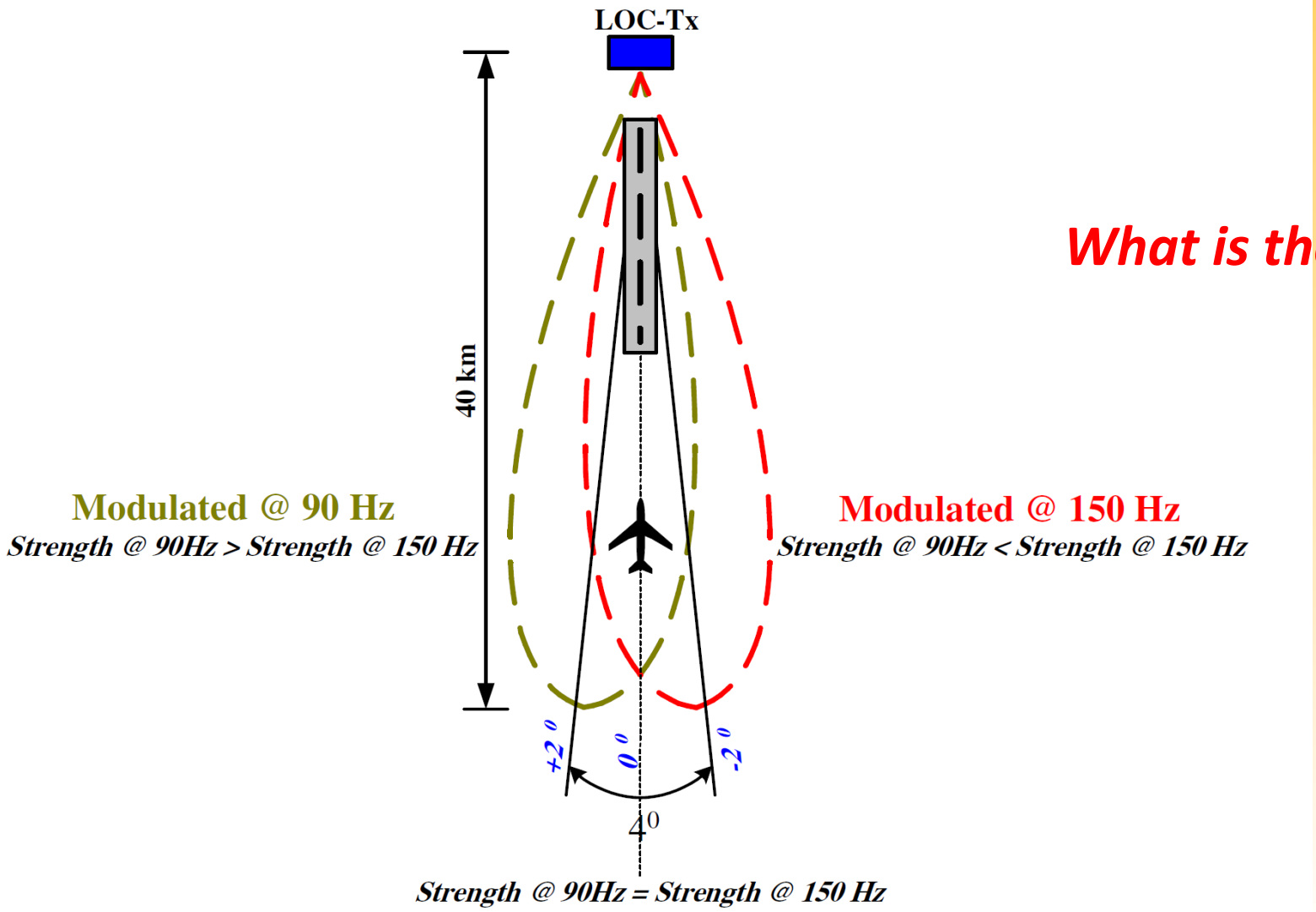
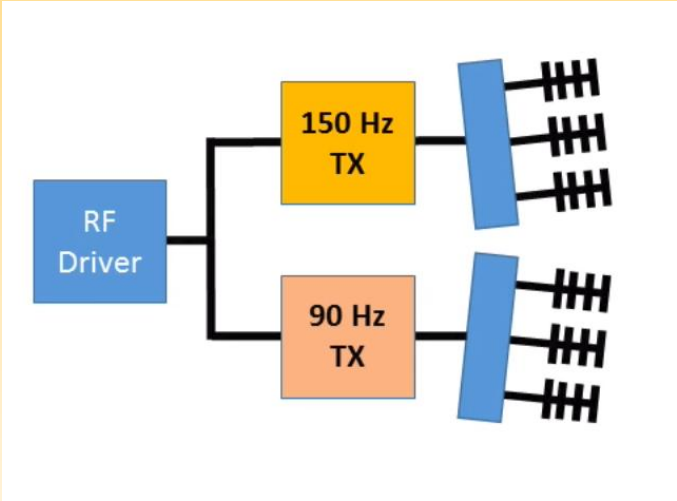
What if we used two group of antennas?

One to the modulated at 150 Hz pointing towards the right side of the runway, and the other modulated at 90 Hz pointing to the left side?



What is the issue?



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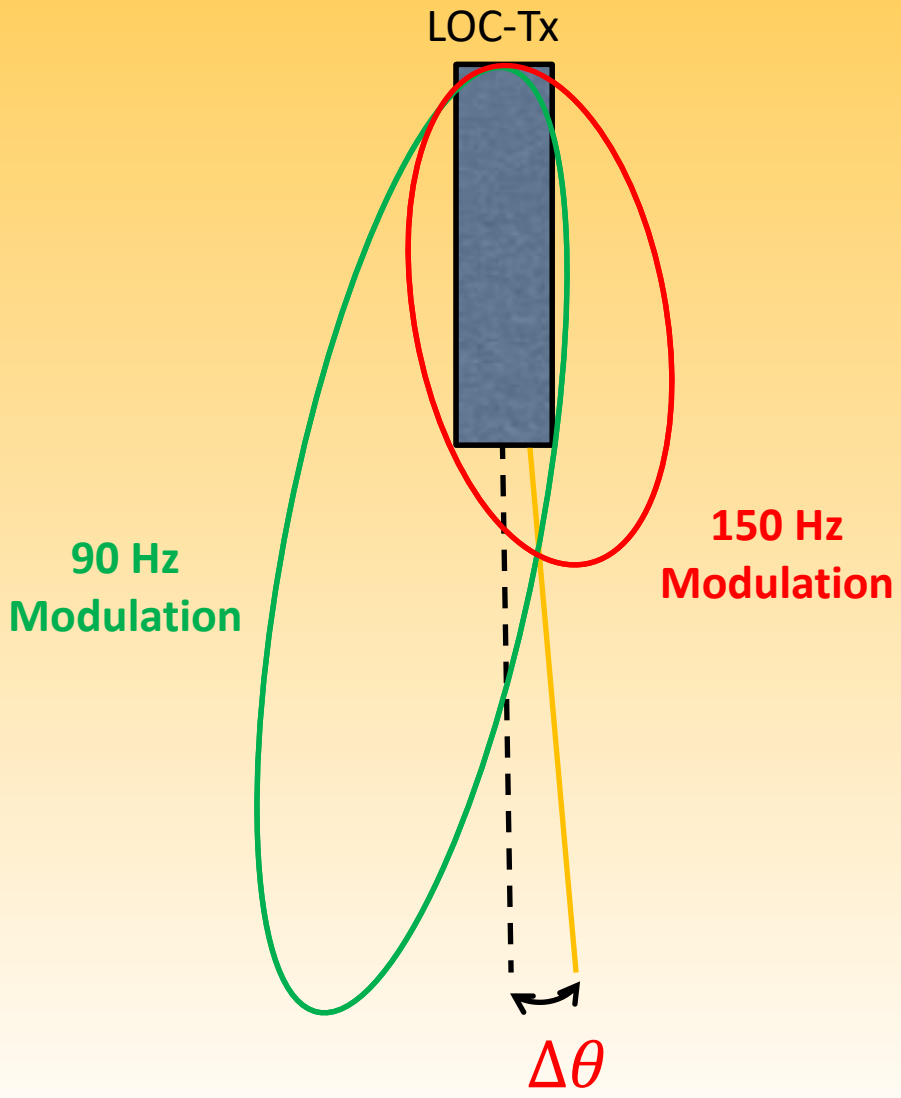
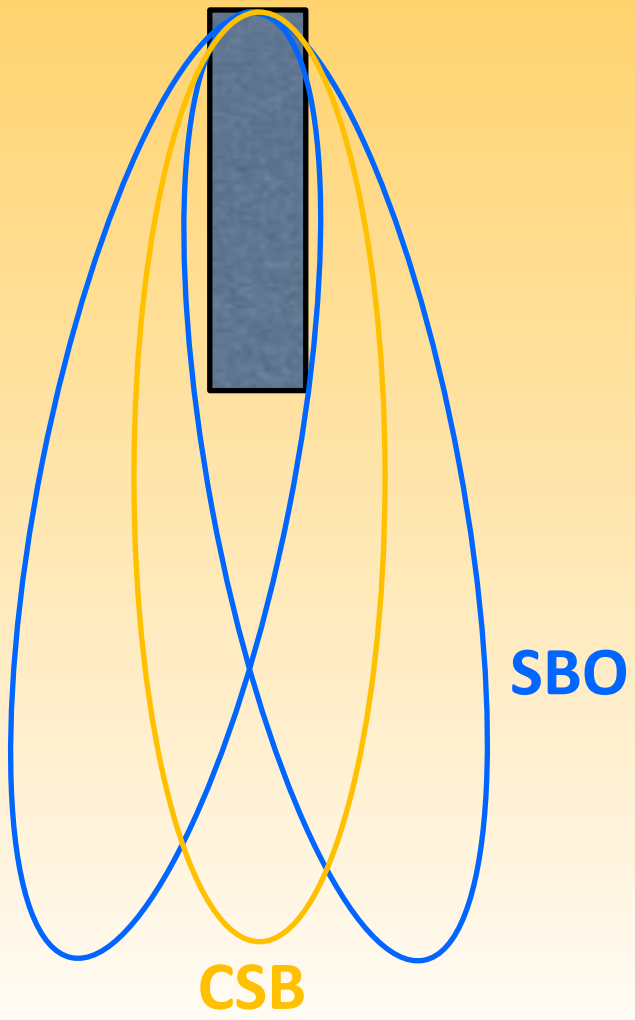
Solution : Used two types of modulation:



*Carrier with Side Band (**CSB**)*
*Side Bands Only (**SBO**)*



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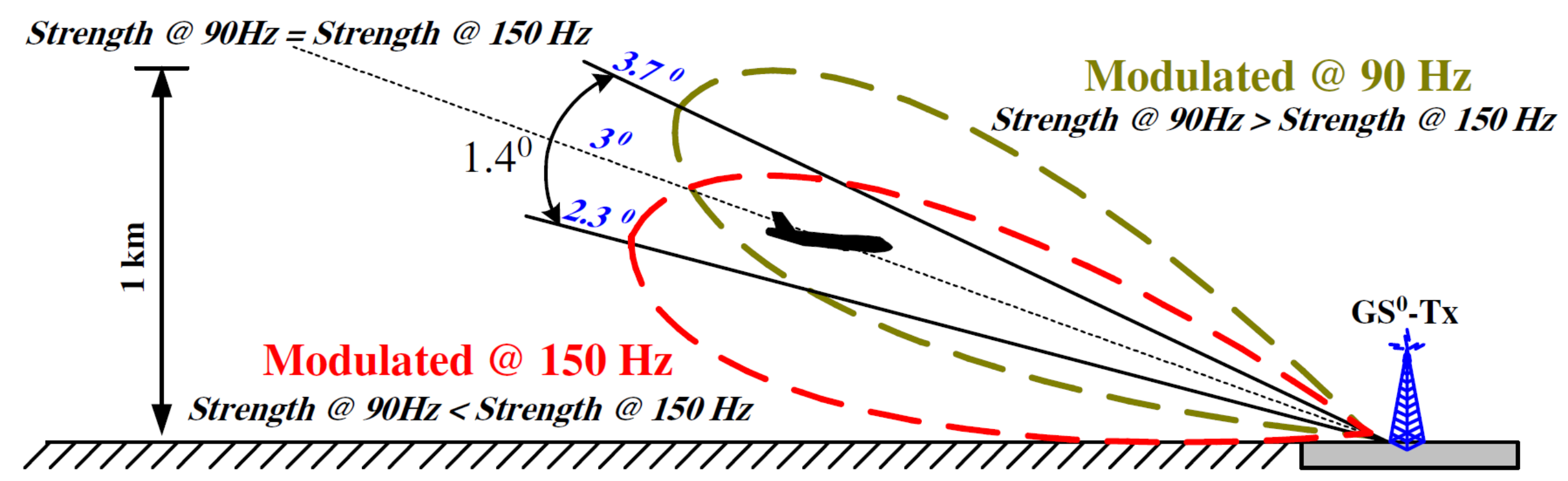
Glideslope:

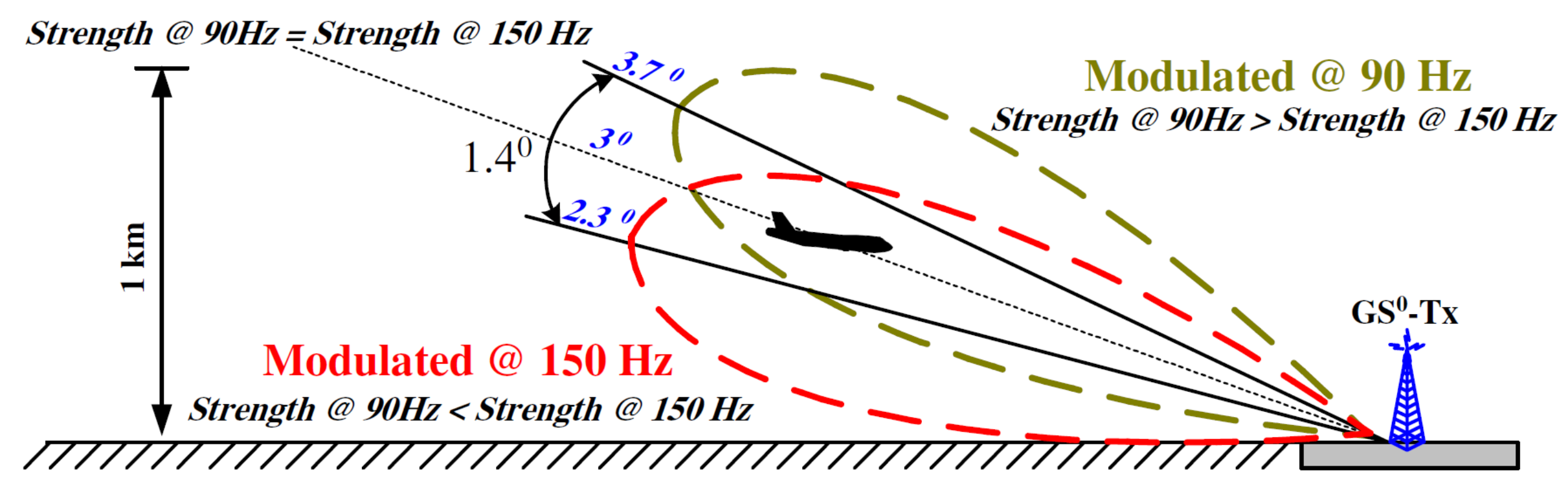
The glideslope systems provides vertical angle information to the pilot, so a fixed descent rate can be achieved.

The principle of operation is very similar to one of the localizer.



Glideslope (contd.):

There is one glideslope transmitter per runway.
Frequency range: UHF 329 – 335 MHz, with 20 channels.
Vertical range: 1 km.
Typical inclination: 3°
Typical deviation: +- 0.7^o (1.4^o total).



Glideslope:

Glideslope signals can be affected by:

- Terrain profile – Moving vehicles – Terrain Moisture - Snow

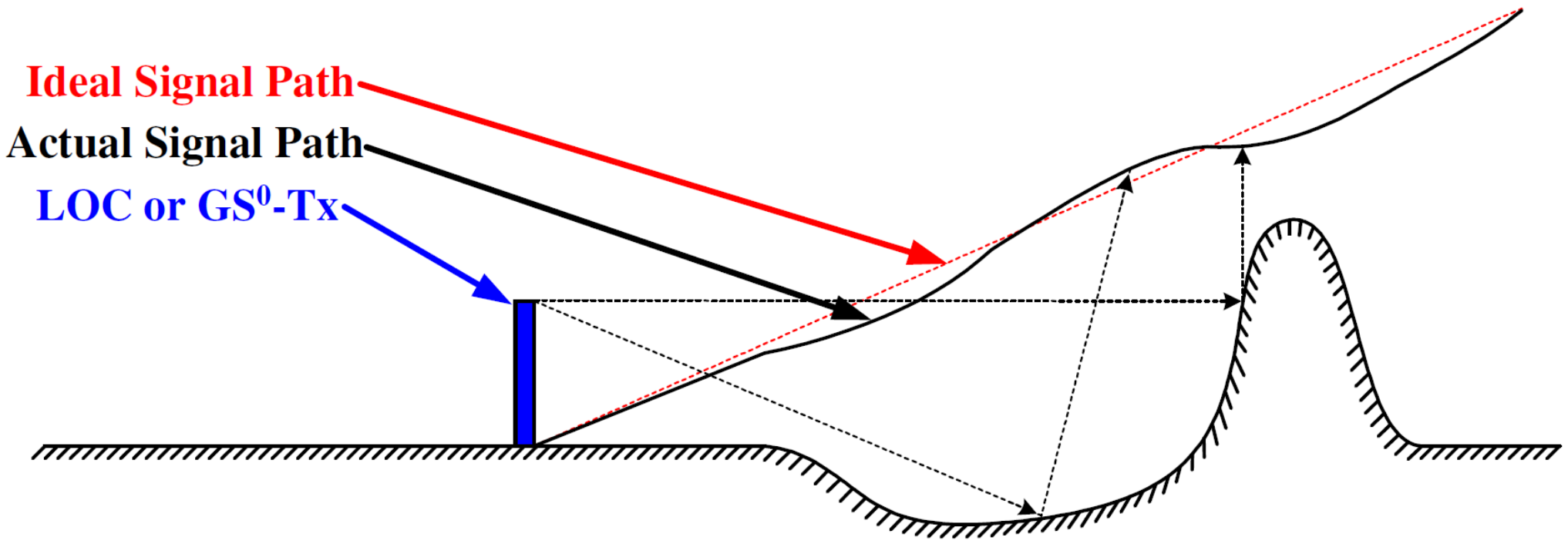


Figure-7.15 Terrain effect in ILS *[K3-30]*

From: Avionics Made Simple by Abdulla, Svoboda, and Rodrigues.

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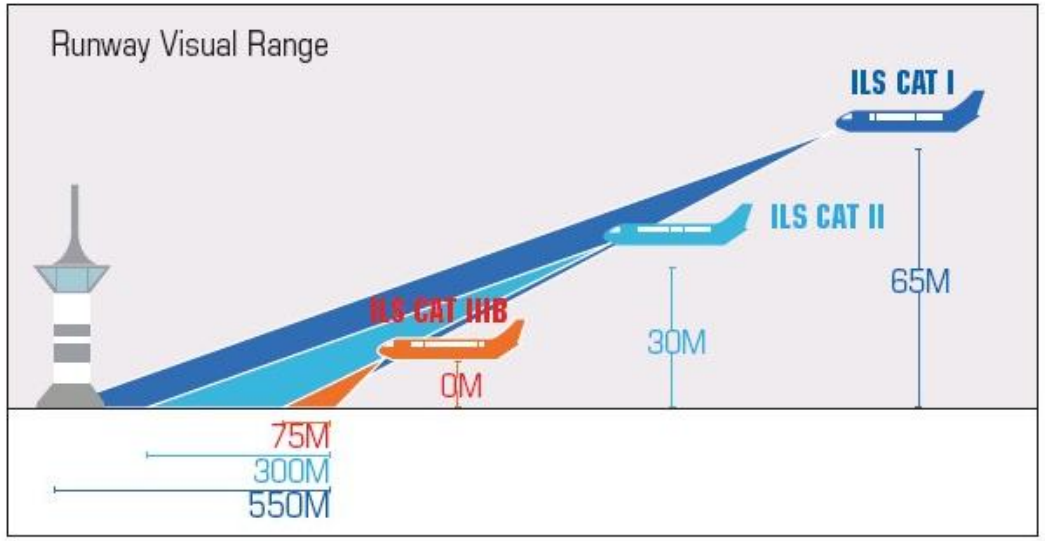
ILS categories

Category	Decision height	RVR (Runway Visual Range)
I	> 200 ft (60m)	> 550 m (1800 ft) or visibility > 800m (2600 ft)
II	30-60m (100-200ft)	ICAO: > 350m (1200ft) FAA: 1200-2400ft (350-800m) JAA(EASA): > 300m (1000ft)
III A	< 100ft (30m)	> 700ft (200m)
III B	< 50ft (15m)	ICAO/FAA: 150-700ft (50-200m) JAA(EASA): 250-700ft (75-200m)



From: Avionics Made Simple by Abdulla, Svoboda, and Rodrigues.

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Category I

Category	Decision height	RVR (Runway Visual Range)
I	> 200 ft (60m)	> 550 m (1800 ft) or visibility > 800m (2600 ft)
II	30-60m (100-200ft)	ICAO: > 350m (1200ft) FAA: 1200-2400ft (350-800m) JAA(EASA): > 300m (1000ft)
III A	< 100ft (30m)	> 700ft (200m)
III B	< 50ft (15m)	ICAO/FAA: 150-700ft (50-200m) JAA(EASA): 250-700ft (75-200m)

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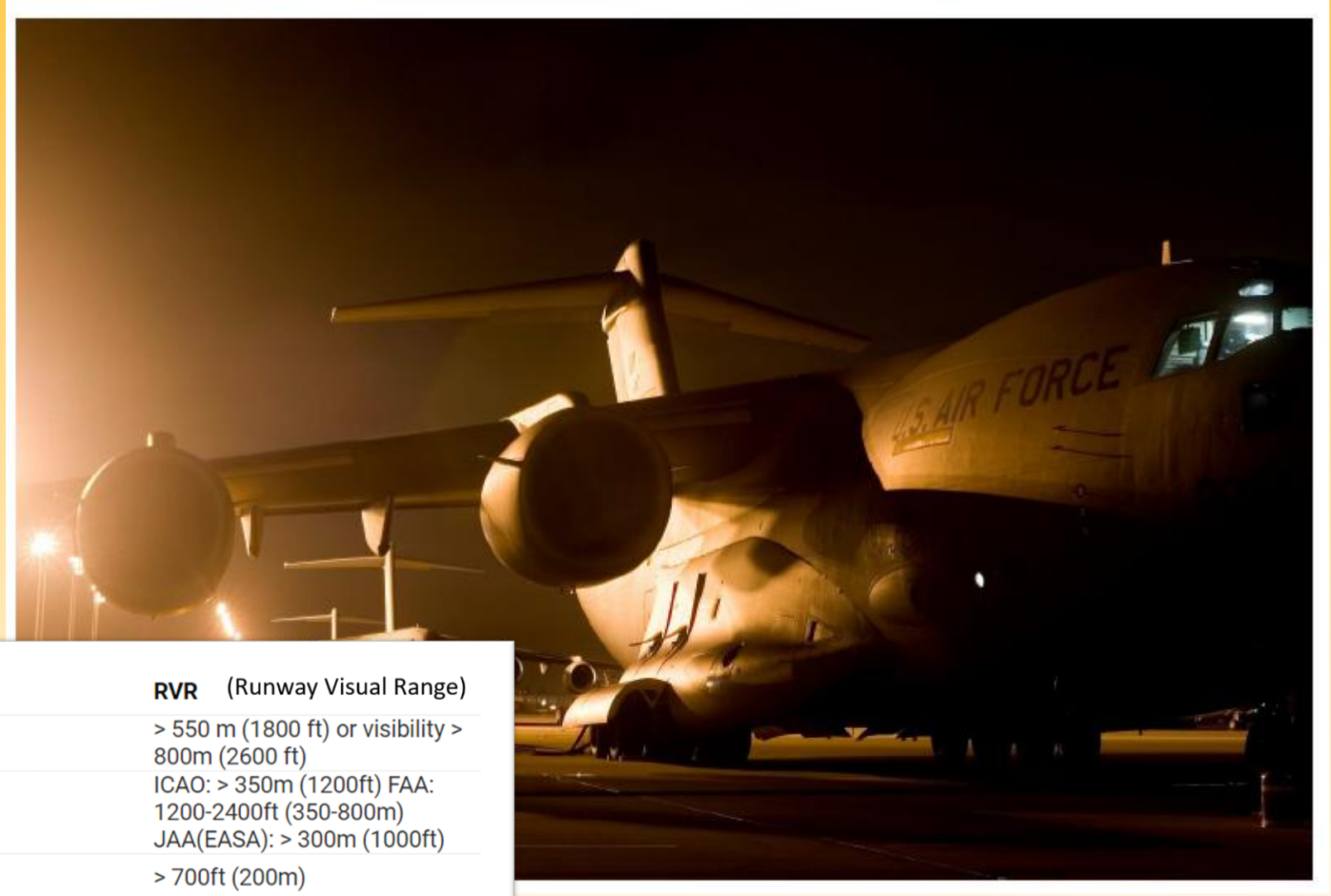
Category II



Category	Decision height	RVR (Runway Visual Range)
I	> 200 ft (60m)	> 550 m (1800 ft) or visibility > 800m (2600 ft)
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III B	< 50ft (15m)	ICAO/FAA: 150-700ft (50-200m) JAA(EASA): 250-700ft (75-200m)

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Category IIIA

Category	Decision height	RVR (Runway Visual Range)
I	> 200 ft (60m)	> 550 m (1800 ft) or visibility > 800m (2600 ft)
II	30-60m (100-200ft)	ICAO: > 350m (1200ft) FAA: 1200-2400ft (350-800m) JAA(EASA): > 300m (1000ft)
III A	< 100ft (30m)	> 700ft (200m)
III B	< 50ft (15m)	ICAO/FAA: 150-700ft (50-200m) JAA(EASA): 250-700ft (75-200m)

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Category IIIB

Category	Decision height	RVR (Runway Visual Range)
I	> 200 ft (60m)	> 550 m (1800 ft) or visibility > 800m (2600 ft)
II	30-60m (100-200ft)	ICAO: > 350m (1200ft) FAA: 1200-2400ft (350-800m) JAA(EASA): > 300m (1000ft)
III A	< 100ft (30m)	> 700ft (200m)
III B	< 50ft (15m)	ICAO/FAA: 150-700ft (50-200m) JAA(EASA): 250-700ft (75-200m)

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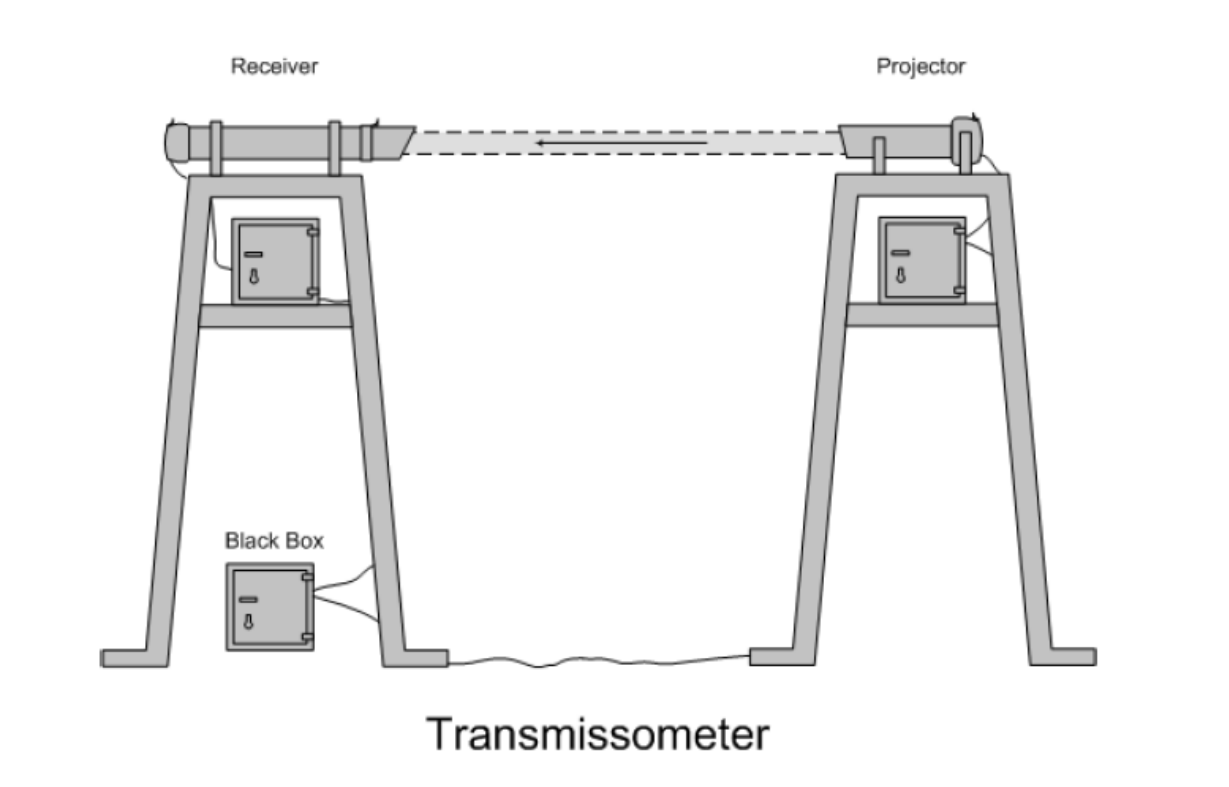
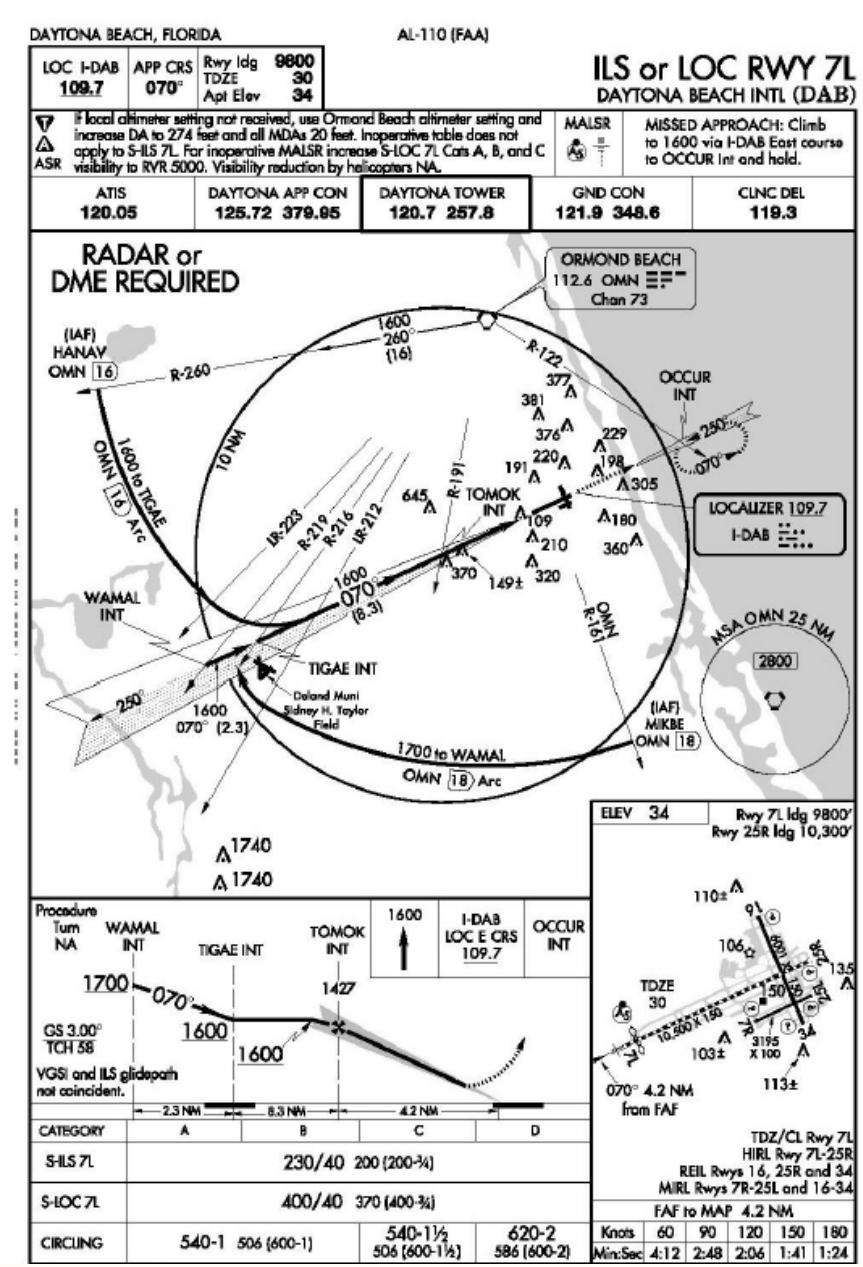
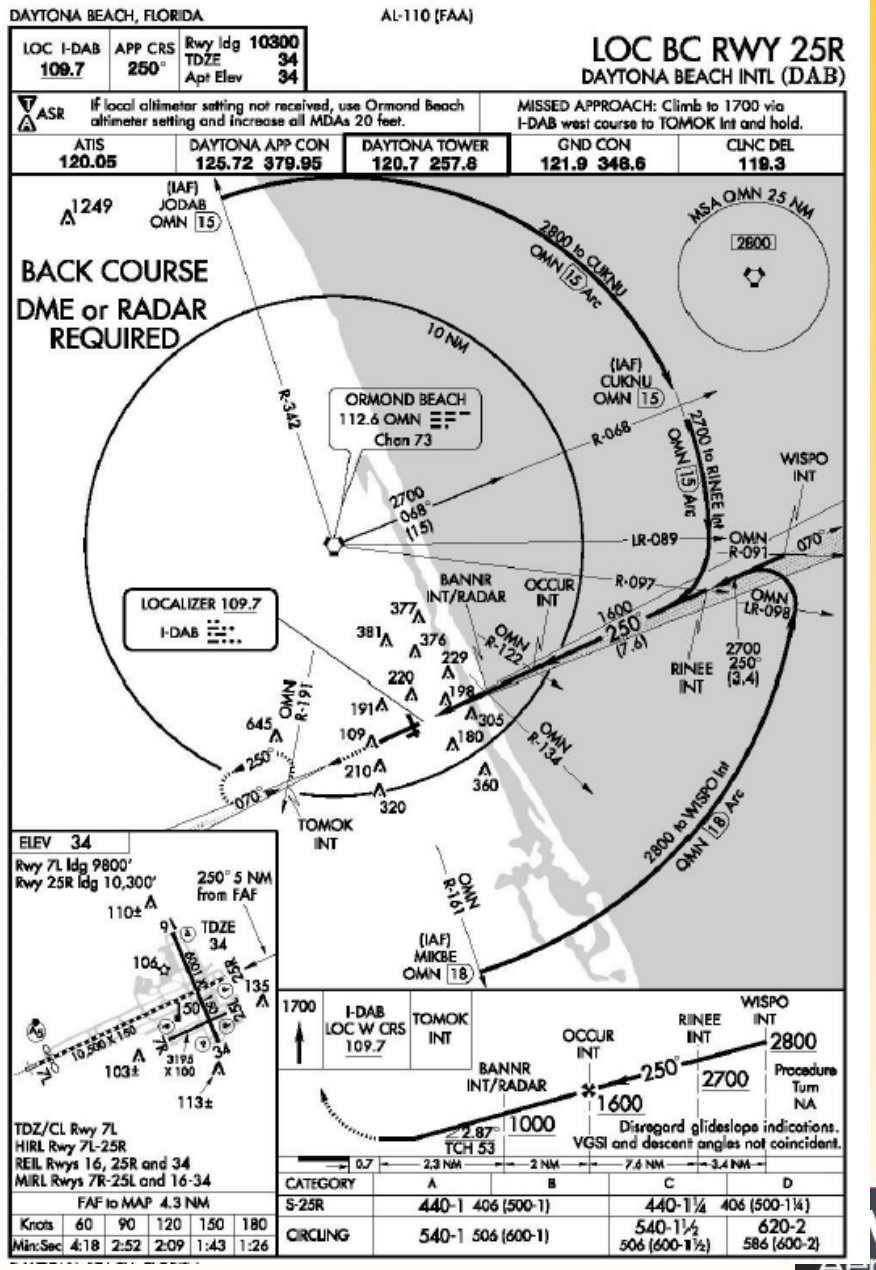
How is runway visibility measured?



Normal and Back course for DAB





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Questions?

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